Program Summary - Riedership_Regression_Covid.sas

Execution Environment

Author:	u64420076
File:	/home/u64420076/BAN000_Personal/Riedership_Regression_Covid.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS01-USW2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	4/17/2026, 11:57:06 AM
Browser Host:	POOL-173-35-226-51.CPE.NET.CABLE.ROGERS.COM
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36
Application Server:	ODAMID00-USW2.ODA.SAS.COM

Code: Riedership_Regression_Covid.sas

```
/* Importr Data */

FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';
PROC IMPORT DATAFILE=REFFILE
    DBMS=CSV
    OUT=WORK.TRANSIT;
    GETNAMES=YES;
RUN;

/* dummy variables */

data transit_reg;
    set transit;

    /* Month dummies: month 12 is reference */
    month_1  = (month = 1);
    month_2  = (month = 2);
    month_3  = (month = 3);
    month_4  = (month = 4);
    month_5  = (month = 5);
    month_6  = (month = 6);
    month_7  = (month = 7);
    month_8  = (month = 8);
    month_9  = (month = 9);
    month_10 = (month = 10);
    month_11 = (month = 11);

    /* COVID period dummies: Pre-COVID is reference */
    covidrec  = (covid_period = "COVID/Rec");
    postcovid = (covid_period = "Post-COVI");
run;

/* create period subsets */
data pre_covid2 ;
    set transit_reg;
    if covid_period = "Pre-COVID" then output pre_covid2;
run;

data covid_rec2;
    set transit_reg;
    if covid_period = "COVID/Rec" then output covid_rec2;
run;

data post_covid2;
    set transit_reg;
    if covid_period = "Post-COVI" then output post_covid2;
run;

/*_____________________________________PRE-COVID TEST MODELS_____________________________________*/

/* covid_rec2 Time Baseline Model */
proc reg data=covid_rec2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
```

```
        / vif tol;
    title "PROC REG Model 1: Baseline";
run; quit;

/* covid_rec2 Lag Baseline Model */
proc reg data=covid_rec2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        ridership_lag1 ridership_lag12
        / vif tol;
    title "PROC REG Model 2: Lag Baseline";
run; quit;

/* covid_rec2 full model */
proc reg data=covid_rec2;
    model ridership=
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol ;
     title 'Pre-Covid PROC REG Model 3.1';

run; quit;

/* step-wise */
proc reg data=covid_rec2;
    model ridership=
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol SELECTION=STEPWISE SLENTRY=0.05 SLSTAY=0.15;
     title 'Pre-Covid PROC REG STEP Model 3.2';

run; quit;

/*_____________________________________________COVID MODELS_____________________________________________*/


/* Month-only Seasonal Baseline */
proc reg data=covid_rec2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        / vif tol;
    title "Pre-COVID Model P1: Month-only Seasonal Baseline";
run; quit;


/* Lag Memory */
proc reg data=covid_rec2;
    model ridership =
        ridership_lag1
        ridership_lag12
        / vif tol;
    title "Pre-COVID Model P2: Lag Memory Model";
run; quit;

/* Reduced External Model */
proc reg data=covid_rec2;
    model ridership =
        ridership_lag1
        ridership_lag12
        gas_price_lag1
        unemployment_rate_lag1
        / vif tol;
    title "Pre-COVID Model P3: Reduced External Model";
run; quit;



proc reg data=covid_rec2;
    model ridership=
        gas_price_lag1 unemployment_rate_lag1
```

```
        avg_temp total_precip_mm total_snow_cm
        / vif tol ;
     title 'Pre-Covid Model P4: Whithout Ridership History';

run; quit;
```

Log: Riedership_Regression_Covid.sas

Notes (27)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69
70         /* Importr Data */
71
72         FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';
73
74         PROC IMPORT DATAFILE=REFFILE
75         DBMS=CSV
76         OUT=WORK.TRANSIT;
77         GETNAMES=YES;
78         RUN;

NOTE: Unable to open parameter catalog: SASUSER.PARMS.PARMS.SLIST in update mode. Temporary parameter values will be saved to
WORK.PARMS.PARMS.SLIST.
79          /*********************************************************************
80          *   PRODUCT:   SAS
81          *   VERSION:   9.4
82          *   CREATOR:   External File Interface
83          *   DATE:      17APR26
84          *   DESC:      Generated SAS Datastep Code
85          *   TEMPLATE SOURCE:  (None Specified.)
86          *********************************************************************/
87             data WORK.TRANSIT    ;
88             %let _EFIERR_ = 0; /* set the ERROR detection macro variable */
89             infile REFFILE delimiter = ',' MISSOVER DSD  firstobs=2 ;
90                info format "Unnamed: 0"N best32. ;
91                info format date yymmdd10. ;
92                info format year best32. ;
93                info format month best32. ;
94                info format quarter best32. ;
95                info format season $6. ;
96                info format covid_period $9. ;
97                info format ridership best32. ;
98                info format gas_price_regular best32. ;
99                info format unemployment_rate best32. ;
100               info format avg_temp best32. ;
101               info format total_precip_mm best32. ;
102               info format total_snow_cm best32. ;
103               info format avg_max_temp best32. ;
104               info format avg_min_temp best32. ;
105               info format climate_data_flag $5. ;
106               info format days_in_month best32. ;
107               info format missing_temp_days best32. ;
108               info format missing_precip_days best32. ;
109               info format missing_snow_days best32. ;
110               info format temp_completeness best32. ;
111               info format precip_completeness best32. ;
112               info format snow_completeness best32. ;
113               info format ridership_lag1 best32. ;
114               info format ridership_lag12 best32. ;
115               info format gas_price_lag1 best32. ;
116               info format unemployment_rate_lag1 best32. ;
117               info format avg_temp_lag1 best32. ;
118               info format total_precip_mm_lag1 best32. ;
119               info format total_snow_cm_lag1 best32. ;
```

```
120             format "Unnamed: 0"N best12. ;
121             format date yymmdd10. ;
122             format year best12. ;
123             format month best12. ;
124             format quarter best12. ;
125             format season $6. ;
126             format covid_period $9. ;
127             format ridership best12. ;
128             format gas_price_regular best12. ;
129             format unemployment_rate best12. ;
130             format avg_temp best12. ;
131             format total_precip_mm best12. ;
132             format total_snow_cm best12. ;
133             format avg_max_temp best12. ;
134             format avg_min_temp best12. ;
135             format climate_data_flag $5. ;
136             format days_in_month best12. ;
137             format missing_temp_days best12. ;
138             format missing_precip_days best12. ;
139             format missing_snow_days best12. ;
140             format temp_completeness best12. ;
141             format precip_completeness best12. ;
142             format snow_completeness best12. ;
143             format ridership_lag1 best12. ;
144             format ridership_lag12 best12. ;
145             format gas_price_lag1 best12. ;
146             format unemployment_rate_lag1 best12. ;
147             format avg_temp_lag1 best12. ;
148             format total_precip_mm_lag1 best12. ;
149             format total_snow_cm_lag1 best12. ;
150          input
151                      "Unnamed: 0"N
152                      date
153                      year
154                      month
155                      quarter
156                      season  $
157                      covid_period  $
158                      ridership
159                      gas_price_regular
160                      unemployment_rate
161                      avg_temp
162                      total_precip_mm
163                      total_snow_cm
164                      avg_max_temp
165                      avg_min_temp
166                      climate_data_flag  $
167                      days_in_month
168                      missing_temp_days
169                      missing_precip_days
170                      missing_snow_days
171                      temp_completeness
172                      precip_completeness
173                      snow_completeness
174                      ridership_lag1
175                      ridership_lag12
176                      gas_price_lag1
177                      unemployment_rate_lag1
178                      avg_temp_lag1
179                      total_precip_mm_lag1
180                      total_snow_cm_lag1
181          ;
182          if _ERROR_ then call symputx('_EFIERR_',1);  /* set ERROR detection macro variable */
183          run;

NOTE: The infile REFFILE is:
      Filename=/home/u64420076/BAN000_Personal/main_dataset_modeling.csv,
      Owner Name=u64420076,Group Name=oda,
      Access Permission=-rw-r--r--,
      Last Modified=16Apr2026:13:26:50,
      File Size (bytes)=38716

NOTE: 179 records were read from the infile REFFILE.
      The minimum record length was 166.
      The maximum record length was 267.
NOTE: The data set WORK.TRANSIT has 179 observations and 30 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              10880.87k
      OS Memory           29984.00k
      Timestamp           04/17/2026 03:57:00 PM
      Step Count                        24  Switch Count  2
      Page Faults                       0
      Page Reclaims                     191
      Page Swaps                        0
      Voluntary Context Switches        16
      Involuntary Context Switches      0
```

```
      Block Input Operations           0
      Block Output Operations          280


179 rows created in WORK.TRANSIT from REFFILE.



NOTE: WORK.TRANSIT data set was successfully created.
NOTE: The data set WORK.TRANSIT has 179 observations and 30 variables.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.07 seconds
      user cpu time       0.05 seconds
      system cpu time     0.02 seconds
      memory              10880.87k
      OS Memory           30244.00k
      Timestamp           04/17/2026 03:57:00 PM
      Step Count                       24  Switch Count  8
      Page Faults                      4
      Page Reclaims                    2930
      Page Swaps                       0
      Voluntary Context Switches       110
      Involuntary Context Switches     1
      Block Input Operations           608
      Block Output Operations          376


184
185        /* dummy variables */
186
187        data transit_reg;
188            set transit;
189
190            /* Month dummies: month 12 is reference */
191            month_1  = (month = 1);
192            month_2  = (month = 2);
193            month_3  = (month = 3);
194            month_4  = (month = 4);
195            month_5  = (month = 5);
196            month_6  = (month = 6);
197            month_7  = (month = 7);
198            month_8  = (month = 8);
199            month_9  = (month = 9);
200            month_10 = (month = 10);
201            month_11 = (month = 11);
202
203            /* COVID period dummies: Pre-COVID is reference */
204            covidrec  = (covid_period = "COVID/Rec");
205            postcovid = (covid_period = "Post-COVI");
206        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT.
NOTE: The data set WORK.TRANSIT_REG has 179 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1060.31k
      OS Memory           24752.00k
      Timestamp           04/17/2026 03:57:00 PM
      Step Count                       25  Switch Count  2
      Page Faults                      0
      Page Reclaims                    115
      Page Swaps                       0
      Voluntary Context Switches       10
      Involuntary Context Switches     0
      Block Input Operations           0
      Block Output Operations          264


207
208        /* create period subsets */
209        data pre_covid2 ;
210            set transit_reg;
211            if covid_period = "Pre-COVID" then output pre_covid2;
212        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.PRE_COVID2 has 109 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1058.68k
      OS Memory           24752.00k
      Timestamp           04/17/2026 03:57:00 PM
      Step Count                       26  Switch Count  2
      Page Faults                      0
      Page Reclaims                    111
```

```
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


213
214        data covid_rec2;
215            set transit_reg;
216            if covid_period = "COVID/Rec" then output covid_rec2;
217        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.COVID_REC2 has 34 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              1058.56k
      OS Memory           24752.00k
      Timestamp           04/17/2026 03:57:00 PM
      Step Count                        27  Switch Count  2
      Page Faults                       0
      Page Reclaims                     117
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


218
219        data post_covid2;
220            set transit_reg;
221            if covid_period = "Post-COVI" then output post_covid2;
222        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.POST_COVID2 has 36 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1060.43k
      OS Memory           24752.00k
      Timestamp           04/17/2026 03:57:00 PM
      Step Count                        28  Switch Count  2
      Page Faults                       0
      Page Reclaims                     117
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           272


223
224        /*_____________________________________PRE-COVID TEST MODELS_____________________________________*/
225
226        /* covid_rec2 Time Baseline Model */
227        proc reg data=covid_rec2;
228            model ridership =
229                month_1 month_2 month_3 month_4 month_5 month_6
230                month_7 month_8 month_9 month_10 month_11
231
232                / vif tol;
233            title "PROC REG Model 1: Baseline";
234        run;

234      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           2.67 seconds
      user cpu time       0.17 seconds
      system cpu time     0.04 seconds
      memory              14891.62k
      OS Memory           37708.00k
      Timestamp           04/17/2026 03:57:03 PM
      Step Count                        29  Switch Count  23
      Page Faults                       0
      Page Reclaims                     15453
      Page Swaps                        0
      Voluntary Context Switches        1037
      Involuntary Context Switches      7
      Block Input Operations            0
      Block Output Operations           1184
```

```
235
236        /* covid_rec2 Lag Baseline Model */
237        proc reg data=covid_rec2;
238            model ridership =
239                month_1 month_2 month_3 month_4 month_5 month_6
240                month_7 month_8 month_9 month_10 month_11
241                ridership_lag1 ridership_lag12
242                / vif tol;
243            title "PROC REG Model 2: Lag Baseline";
244        run;

244      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.68 seconds
      user cpu time       0.24 seconds
      system cpu time     0.04 seconds
      memory              11435.15k
      OS Memory           43240.00k
      Timestamp           04/17/2026 03:57:04 PM
      Step Count                        30  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12706
      Page Swaps                        0
      Voluntary Context Switches        993
      Involuntary Context Switches      13
      Block Input Operations            0
      Block Output Operations           1456

245
246        /* covid_rec2 full model */
247        proc reg data=covid_rec2;
248        model ridership=
249        month_1 month_2 month_3 month_4 month_5 month_6
250                month_7 month_8 month_9 month_10 month_11
251                ridership_lag1 ridership_lag12
252                gas_price_lag1 unemployment_rate_lag1
253                avg_temp total_precip_mm total_snow_cm
254                / vif tol ;
255             title 'Pre-Covid PROC REG Model 3.1';
256
257        run;

257      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.50 seconds
      user cpu time       0.19 seconds
      system cpu time     0.03 seconds
      memory              11222.84k
      OS Memory           43240.00k
      Timestamp           04/17/2026 03:57:04 PM
      Step Count                        31  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12527
      Page Swaps                        0
      Voluntary Context Switches        1069
      Involuntary Context Switches      9
      Block Input Operations            0
      Block Output Operations           1512


258
259        /* step-wise */
260        proc reg data=covid_rec2;
261        model ridership=
262        month_1 month_2 month_3 month_4 month_5 month_6
263                month_7 month_8 month_9 month_10 month_11
264                covidrec postcovid
265                ridership_lag1 ridership_lag12
266                gas_price_lag1 unemployment_rate_lag1
267                avg_temp total_precip_mm total_snow_cm
268                / vif tol SELECTION=STEPWISE SLENTRY=0.05 SLSTAY=0.15;
269             title 'Pre-Covid PROC REG STEP Model 3.2';
270
271        run;

271      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.35 seconds
      user cpu time       0.20 seconds
      system cpu time     0.02 seconds
      memory              11277.62k
      OS Memory           43240.00k
      Timestamp           04/17/2026 03:57:04 PM
      Step Count                        32  Switch Count  23
      Page Faults                       0
```

```
      Page Reclaims                    12088
      Page Swaps                       0
      Voluntary Context Switches       720
      Involuntary Context Switches     4
      Block Input Operations           0
      Block Output Operations          792


272
273        /*_________________________________________COVID MODELS_________________________________________*/
274
275
276        /* Month-only Seasonal Baseline */
277        proc reg data=covid_rec2;
278            model ridership =
279                month_1 month_2 month_3 month_4 month_5 month_6
280                month_7 month_8 month_9 month_10 month_11
281                / vif tol;
282            title "Pre-COVID Model P1: Month-only Seasonal Baseline";
283        run;

283      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.33 seconds
      user cpu time       0.15 seconds
      system cpu time     0.03 seconds
      memory              11234.37k
      OS Memory           42984.00k
      Timestamp           04/17/2026 03:57:05 PM
      Step Count                       33  Switch Count  23
      Page Faults                      0
      Page Reclaims                    12310
      Page Swaps                       0
      Voluntary Context Switches       894
      Involuntary Context Switches     5
      Block Input Operations           0
      Block Output Operations          888


284
285
286        /* Lag Memory */
287        proc reg data=covid_rec2;
288            model ridership =
289                ridership_lag1
290                ridership_lag12
291                / vif tol;
292            title "Pre-COVID Model P2: Lag Memory Model";
293        run;

293      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.25 seconds
      user cpu time       0.11 seconds
      system cpu time     0.02 seconds
      memory              11234.71k
      OS Memory           42984.00k
      Timestamp           04/17/2026 03:57:05 PM
      Step Count                       34  Switch Count  23
      Page Faults                      0
      Page Reclaims                    12011
      Page Swaps                       0
      Voluntary Context Switches       690
      Involuntary Context Switches     4
      Block Input Operations           0
      Block Output Operations          720


294
295        /* Reduced External Model */
296        proc reg data=covid_rec2;
297            model ridership =
298                ridership_lag1
299                ridership_lag12
300                gas_price_lag1
301                unemployment_rate_lag1
302                / vif tol;
303            title "Pre-COVID Model P3: Reduced External Model";
304        run;

304      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.27 seconds
      user cpu time       0.13 seconds
      system cpu time     0.04 seconds
      memory              11201.31k
```

```
OS Memory           42984.00k
Timestamp           04/17/2026 03:57:05 PM
Step Count                        35  Switch Count  23
Page Faults                       0
Page Reclaims                     12067
Page Swaps                        0
Voluntary Context Switches        721
Involuntary Context Switches      13
Block Input Operations            0
Block Output Operations           760


305
306
307
308        proc reg data=covid_rec2;
309        model ridership=
310                gas_price_lag1 unemployment_rate_lag1
311                avg_temp total_precip_mm total_snow_cm
312                / vif tol ;
313             title 'Pre-Covid Model P4: Whithout Ridership History';
314
315        run;

315      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.25 seconds
      user cpu time       0.11 seconds
      system cpu time     0.02 seconds
      memory              11267.59k
      OS Memory           42984.00k
      Timestamp           04/17/2026 03:57:06 PM
      Step Count                        36  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12067
      Page Swaps                        0
      Voluntary Context Switches        714
      Involuntary Context Switches      7
      Block Input Operations            0
      Block Output Operations           760


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337
338        OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
348
```

Results: Riedership_Regression_Covid.sas

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	5.662673E11	51478845811	0.59	0.8178
Error	22	1.923675E12	87439780303		
Corrected Total	33	2.489942E12			

Root MSE	295702	R-Square	0.2274
Dependent Mean	715471	Adj R-Sq	-0.1589

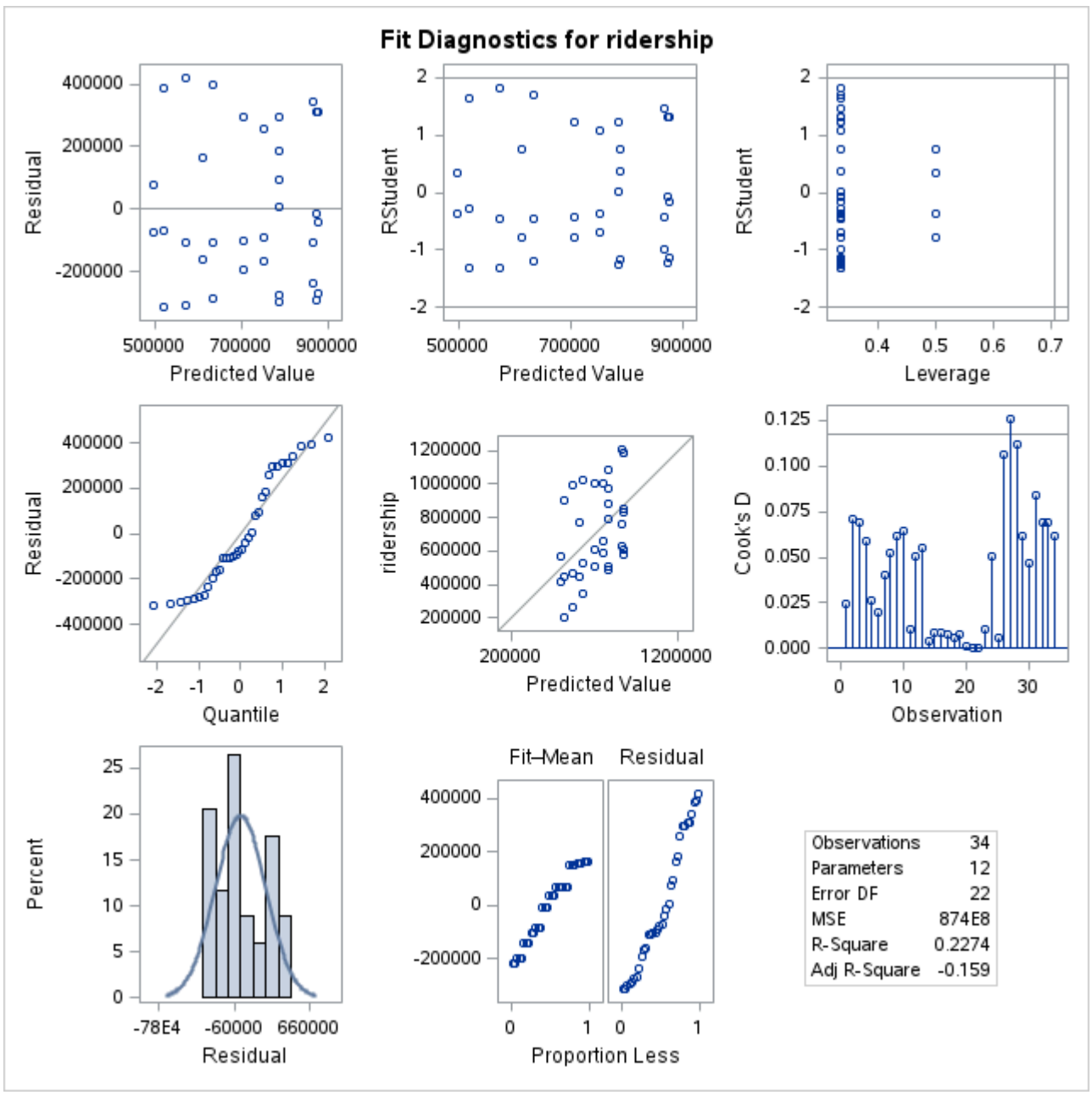
Coeff Var	41.32975		

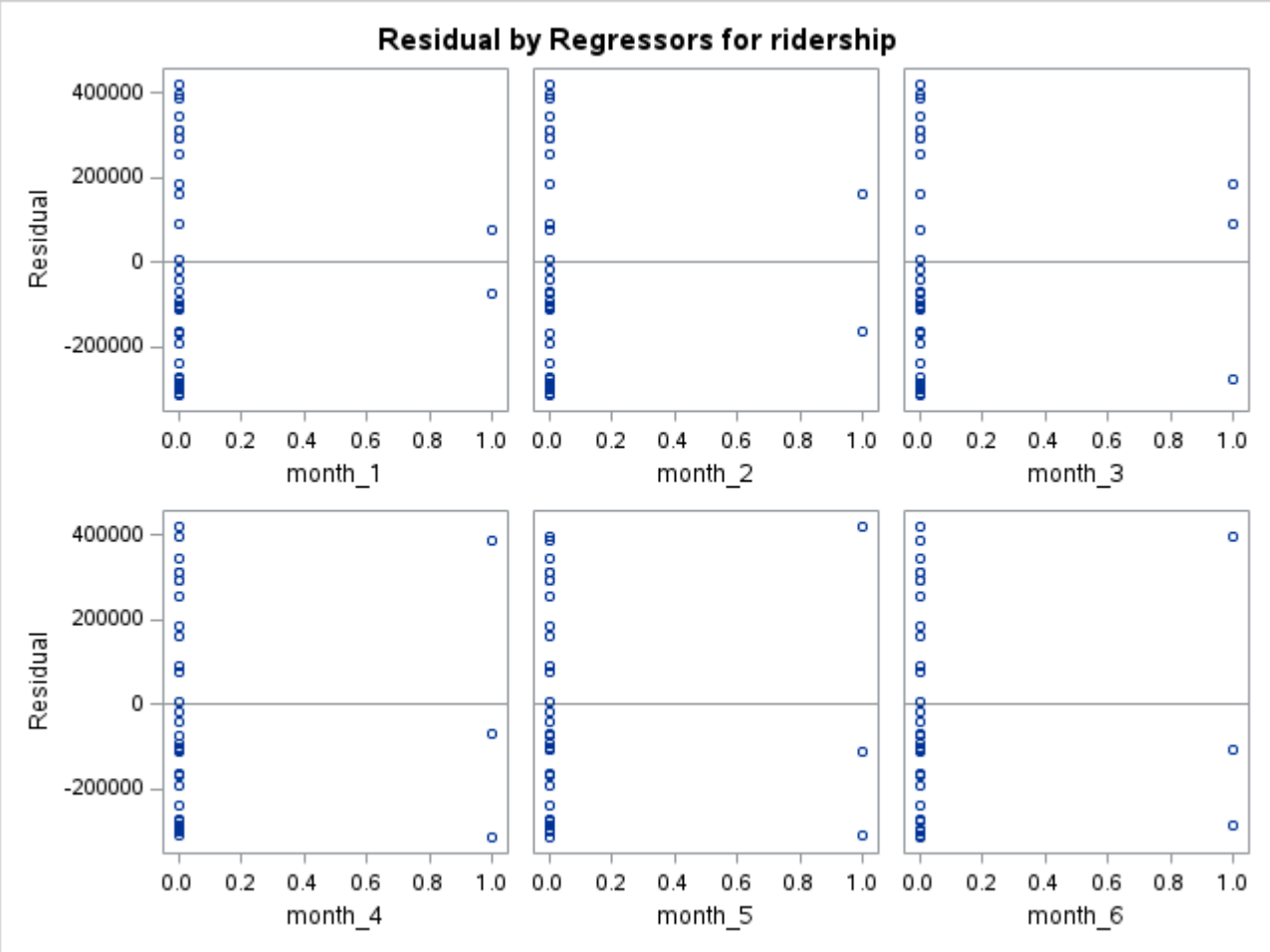
Parameter Estimates

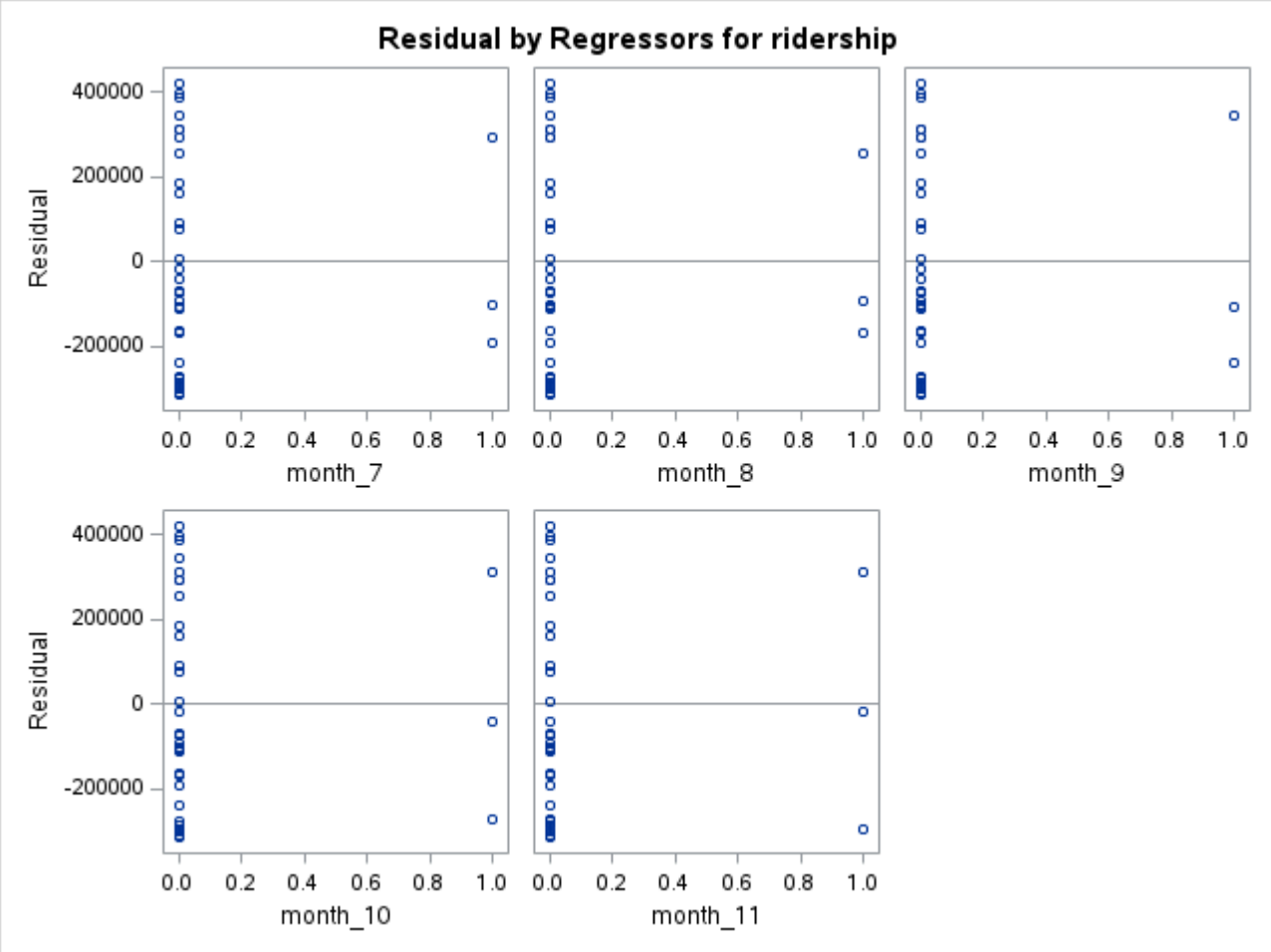
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	785333	170724	4.60	0.0001	.	0
month_1	1	-288833	269938	-1.07	0.2962	0.63750	1.56863
month_2	1	-172333	269938	-0.64	0.5298	0.63750	1.56863
month_3	1	1000.00000	241440	0.00	0.9967	0.54839	1.82353
month_4	1	-265667	241440	-1.10	0.2831	0.54839	1.82353
month_5	1	-212000	241440	-0.88	0.3894	0.54839	1.82353
month_6	1	-152667	241440	-0.63	0.5337	0.54839	1.82353
month_7	1	-80000	241440	-0.33	0.7435	0.54839	1.82353
month_8	1	-33333	241440	-0.14	0.8914	0.54839	1.82353
month_9	1	80000	241440	0.33	0.7435	0.54839	1.82353
month_10	1	91667	241440	0.38	0.7078	0.54839	1.82353
month_11	1	86667	241440	0.36	0.7230	0.54839	1.82353

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership







PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	2.102852E12	1.617578E11	8.36	<.0001
Error	20	3.870905E11	19354526749		
Corrected Total	33	2.489942E12			

Root MSE	139121	**R-Square**	0.8445
Dependent Mean	715471	**Adj R-Sq**	0.7435
Coeff Var	19.44462		

Parameter Estimates

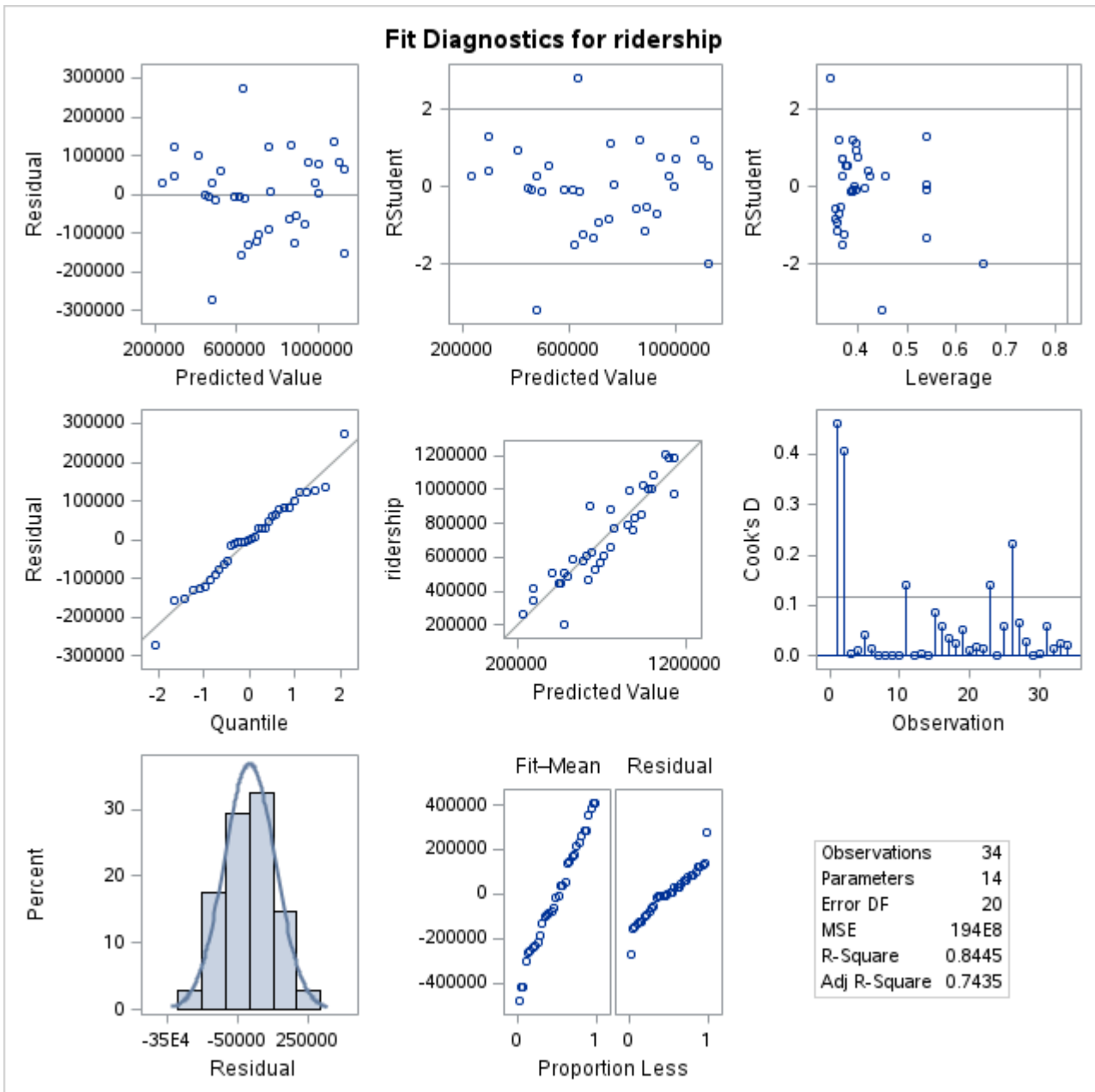
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	421285	126517	3.33	0.0033	.	0
month_1	1	-131875	128589	-1.03	0.3173	0.62183	1.60814
month_2	1	77468	131063	0.59	0.5611	0.59857	1.67064
month_3	1	-47165	114258	-0.41	0.6841	0.54201	1.84499

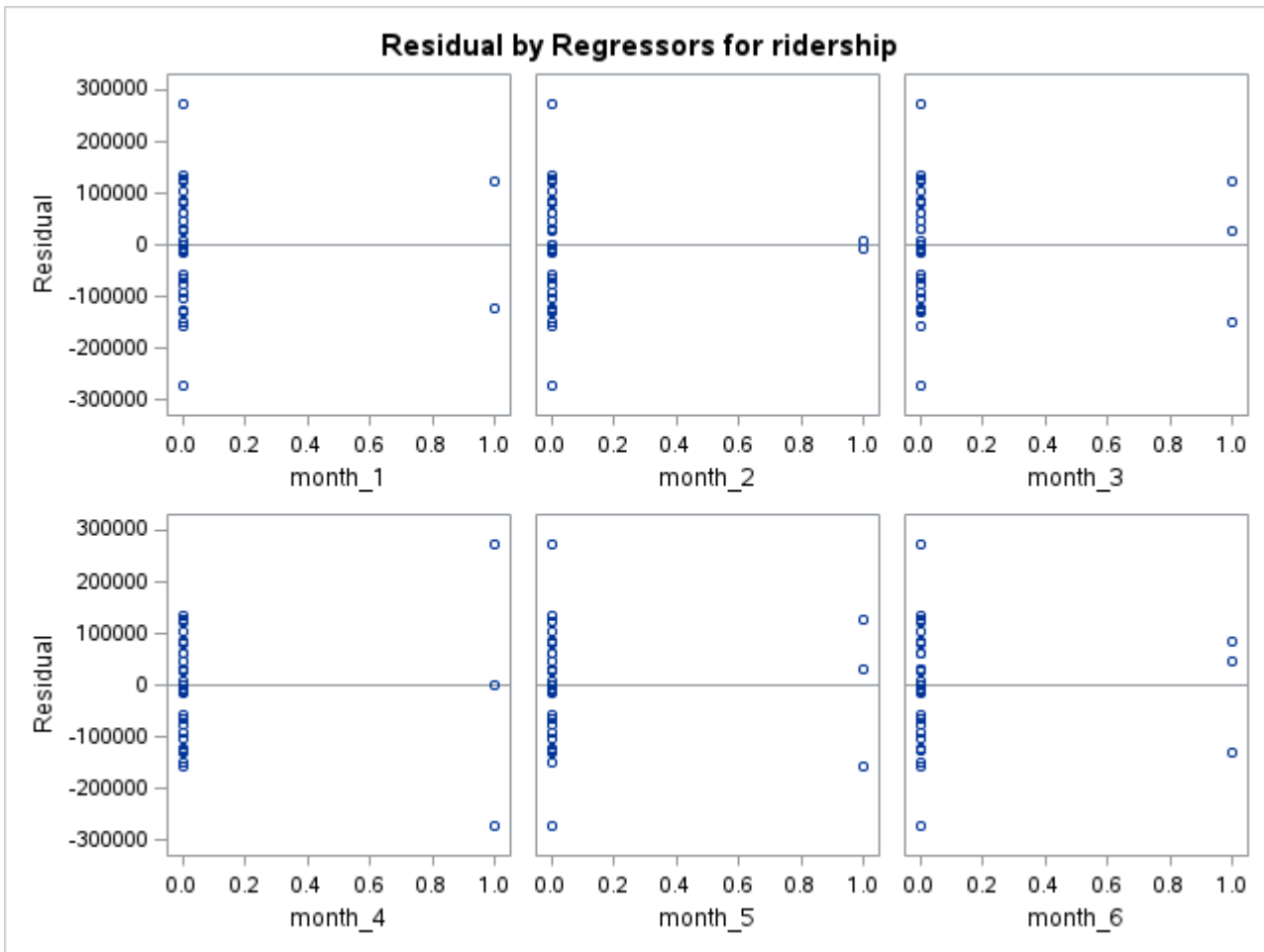
Parameter Estimates

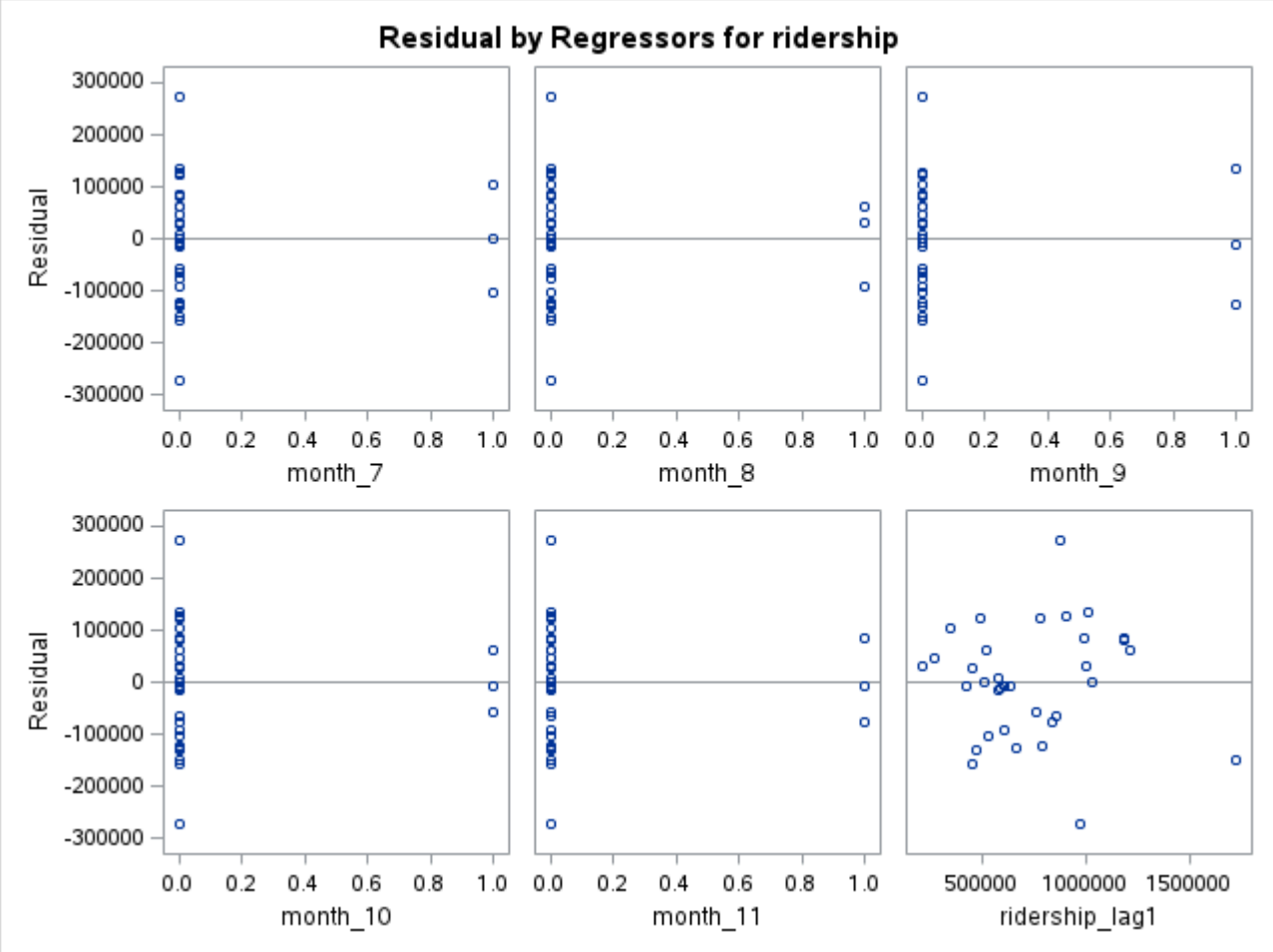
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
month_4	1	-243939	114255	-2.14	0.0453	0.54204	1.84488
month_5	1	-24867	118495	-0.21	0.8359	0.50394	1.98437
month_6	1	12505	116964	0.11	0.9159	0.51722	1.93342
month_7	1	57795	115670	0.50	0.6228	0.52886	1.89087
month_8	1	66240	114566	0.58	0.5696	0.53910	1.85495
month_9	1	170888	114077	1.50	0.1498	0.54373	1.83916
month_10	1	116062	113711	1.02	0.3196	0.54723	1.82737
month_11	1	105150	113738	0.92	0.3663	0.54698	1.82823
ridership_lag1	1	0.60427	0.08899	6.79	<.0001	0.72613	1.37716
ridership_lag12	1	-0.17067	0.04472	-3.82	0.0011	0.89840	1.11309

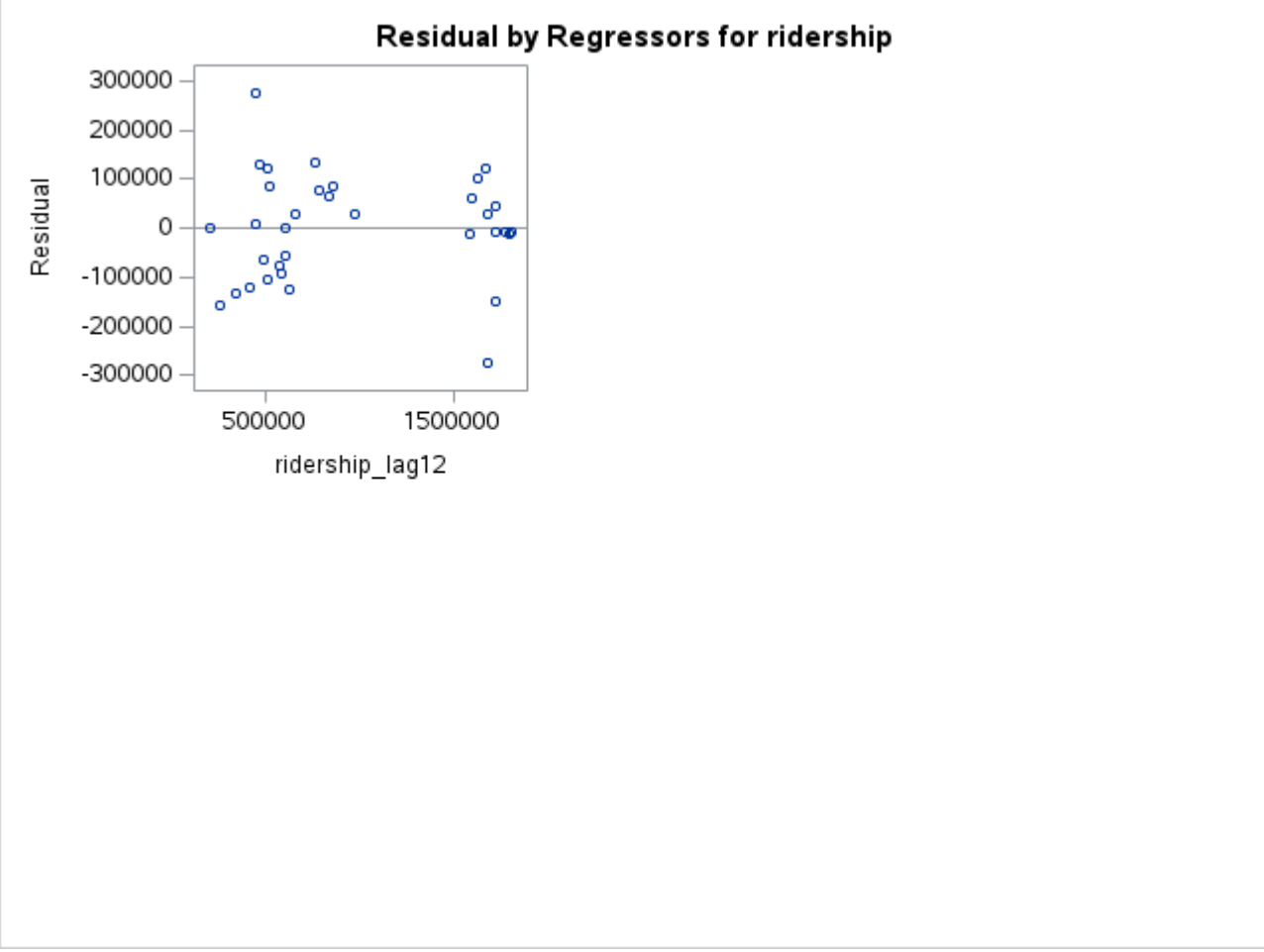
PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership









Pre-Covid PROC REG Model 3.1

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	18	2.334633E12	1.297018E11	12.53	<.0001
Error	15	1.553095E11	10353968796		
Corrected Total	33	2.489942E12			

Root MSE	101754	R-Square	0.9376
Dependent Mean	715471	Adj R-Sq	0.8628
Coeff Var	14.22203		

Parameter Estimates

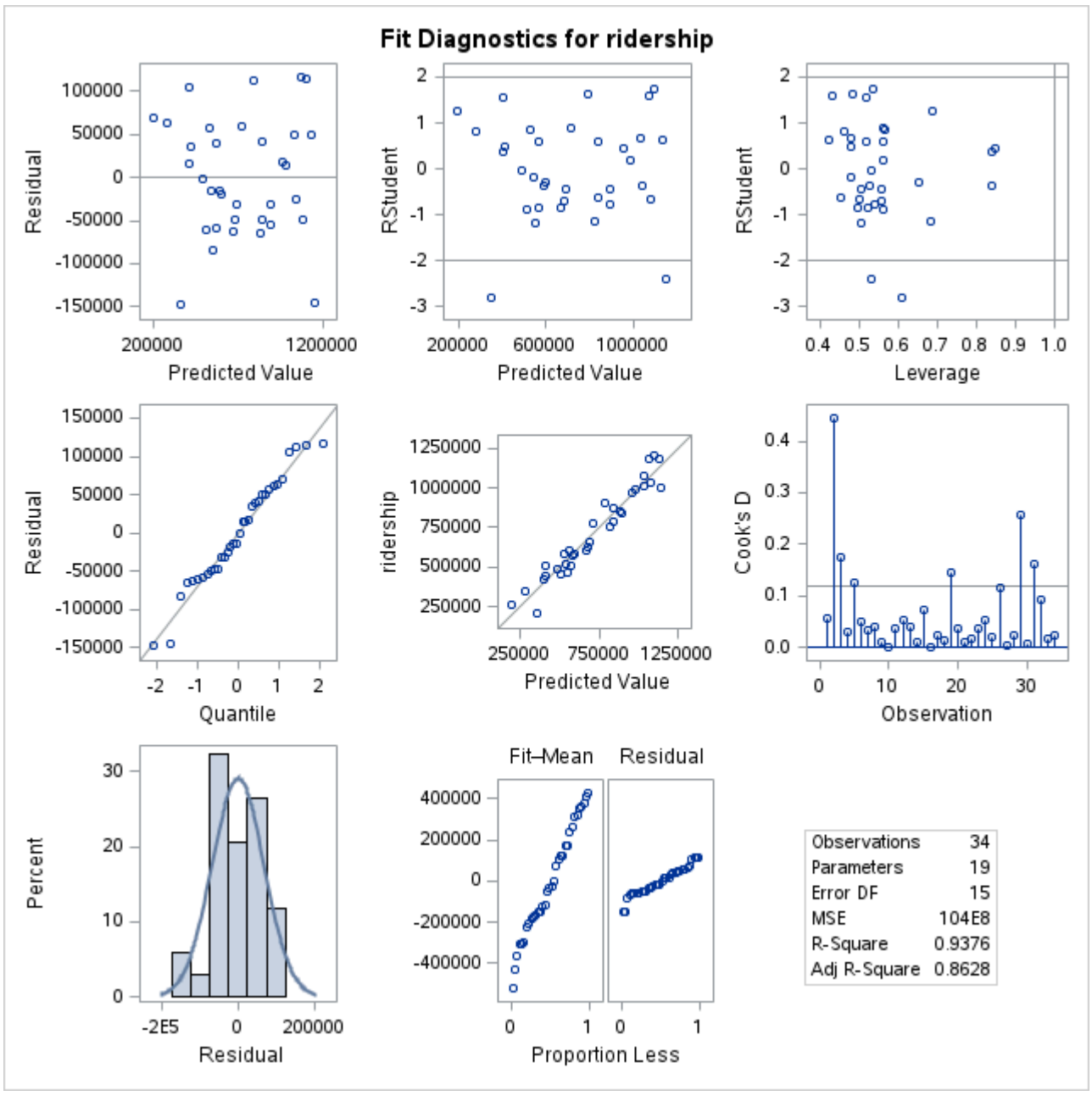
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	-402100	310843	-1.29	0.2154	.	0
month_1	1	-71813	129016	-0.56	0.5860	0.33046	3.02607
month_2	1	37835	121686	0.31	0.7601	0.37147	2.69202
month_3	1	-39792	93799	-0.42	0.6774	0.43023	2.32432

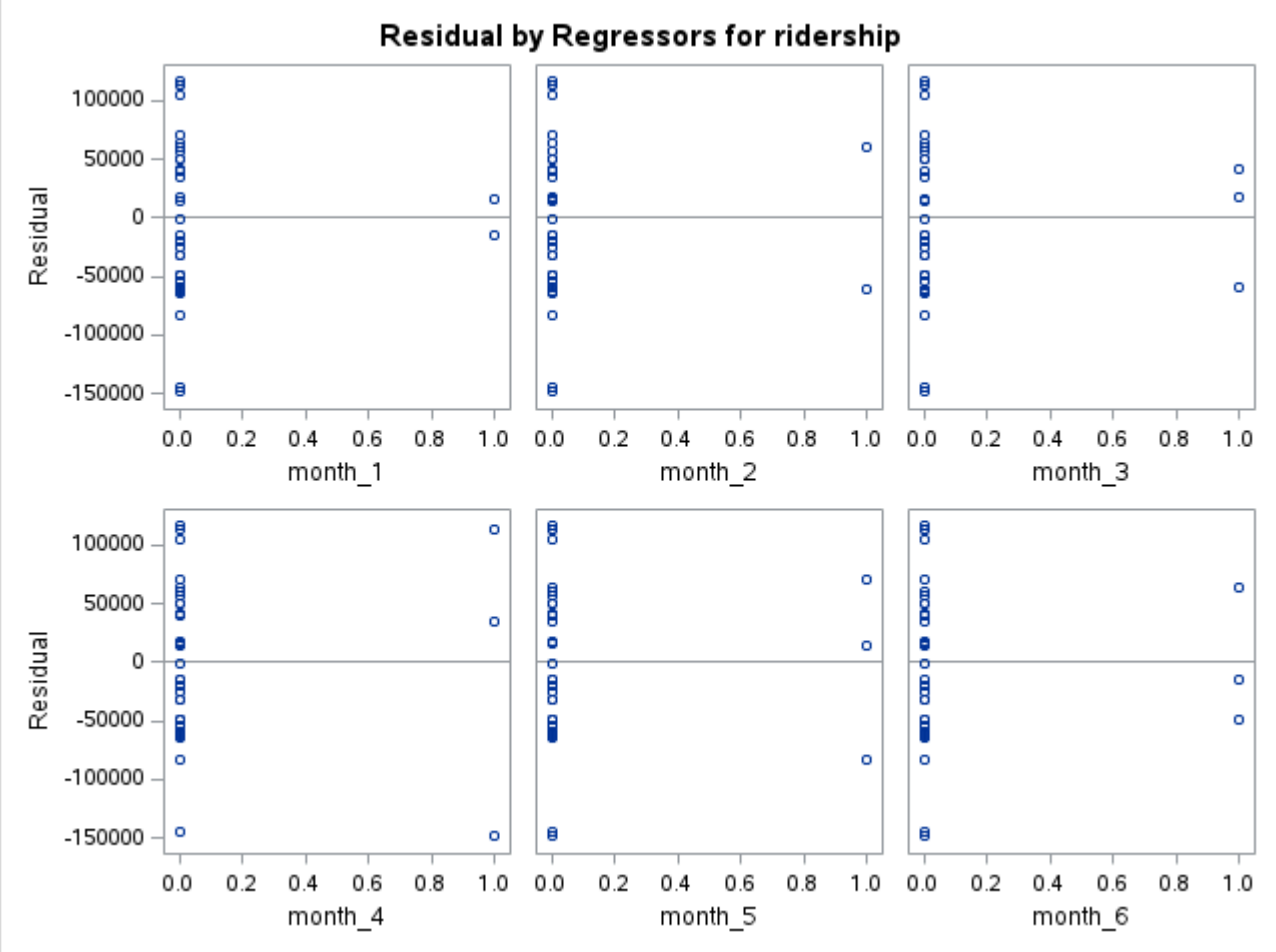
Parameter Estimates

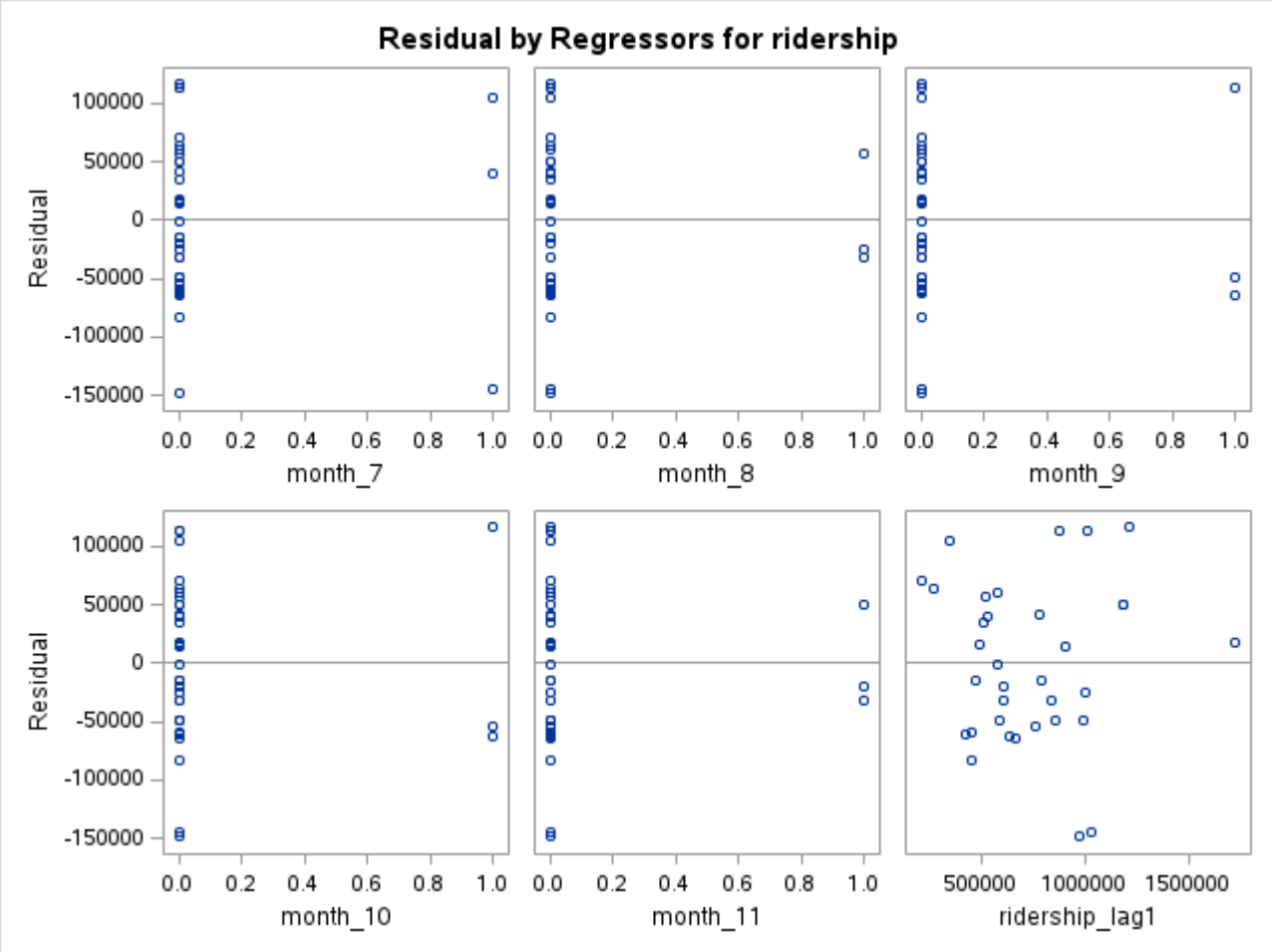
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
month_4	1	-255981	130731	-1.96	0.0691	0.22148	4.51499
month_5	1	-144451	204542	-0.71	0.4909	0.09048	11.05253
month_6	1	-241332	285939	-0.84	0.4119	0.04630	21.59951
month_7	1	-256506	325483	-0.79	0.4429	0.03573	27.98690
month_8	1	-202869	319007	-0.64	0.5344	0.03720	26.88426
month_9	1	-34865	246549	-0.14	0.8894	0.06227	16.05842
month_10	1	17402	166762	0.10	0.9183	0.13612	7.34669
month_11	1	31367	110084	0.28	0.7796	0.31236	3.20146
ridership_lag1	1	0.38000	0.12991	2.93	0.0104	0.18228	5.48608
ridership_lag12	1	0.02031	0.05801	0.35	0.7312	0.28558	3.50159
gas_price_lag1	1	5726.39659	1237.80584	4.63	0.0003	0.19702	5.07572
unemployment_rate_lag1	1	10407	15468	0.67	0.5113	0.17895	5.58821
avg_temp	1	7310.10411	14651	0.50	0.6250	0.01599	62.53039
total_precip_mm	1	-86.59319	786.43015	-0.11	0.9138	0.61352	1.62994
total_snow_cm	1	-901.55507	2996.13245	-0.30	0.7676	0.20692	4.83274

Pre-Covid PROC REG Model 3.1

The REG Procedure
Model: MODEL1
Dependent Variable: ridership




Residual by Regressors for ridership
Residual
100000
50000
0
-50000
-100000
-150000
0.0 0.2 0.4 0.6 0.8 1.0
month_1
month_2
month_3
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
100000
50000
0
-50000
-100000
-150000
0.0 0.2 0.4 0.6 0.8 1.0
month_7
month_8
month_9
month_10
month_11
500000 1000000 1500000
ridership_lag1

Residual by Regressors for ridership
Residual
100000
50000
0
-50000
-100000
-150000
500000 1000000 1500000
ridership_lag12
75 100 125 150 175 200
gas_price_lag1
6 8 10 12 14
unemployment_rate_lag1
-10 0 10 20
avg_temp
0 50 100 150
total_precip_mm
0 10 20 30 40 50
total_snow_cm

Pre-Covid PROC REG STEP Model 3.2

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

Stepwise Selection: Step 1

Variable gas_price_lag1 Entered: R-Square = 0.6685 and C(p) = 49.7122

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1.664605E12	1.664605E12	64.54	<.0001
Error	32	8.253374E11	25791792466		
Corrected Total	33	2.489942E12			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-214067	118938	83548641154	3.24	0.0813
gas_price_lag1	6966.35617	867.14321	1.664605E12	64.54	<.0001

Bounds on condition number: 1, 1

Stepwise Selection: Step 2

Variable ridership_lag1 Entered: R-Square = 0.8200 and C(p) = 15.2762

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	2.041862E12	1.020931E12	70.63	<.0001
Error	31	4.480802E11	14454200807		
Corrected Total	33	2.489942E12			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-222831	89055	90496690327	6.26	0.0178
ridership_lag1	0.39745	0.07780	3.772571E11	26.10	<.0001
gas_price_lag1	4844.28003	770.67193	5.711017E11	39.51	<.0001

Bounds on condition number: 1.4094, 5.6377

Stepwise Selection: Step 3

Variable month_4 Entered: R-Square = 0.8675 and C(p) = 5.8609

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	2.160056E12	7.200187E11	65.48	<.0001
Error	30	3.298864E11	10996212825		
Corrected Total	33	2.489942E12			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-190694	78291	65236954933	5.93	0.0210
month_4	-209202	63810	1.181938E11	10.75	0.0026
ridership_lag1	0.41922	0.06818	4.157293E11	37.81	<.0001
gas_price_lag1	4621.95983	675.60491	5.146475E11	46.80	<.0001

Bounds on condition number: 1.4238, 11.579

Stepwise Selection: Step 4

Variable month_9 Entered: R-Square = 0.8849 and C(p) = 3.6758

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	2.203388E12	5.50847E11	55.75	<.0001
Error	29	2.865544E11	9881184750		
Corrected Total	33	2.489942E12			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-200578	74366	71884010786	7.27	0.0115
month_4	-196866	60775	1.036827E11	10.49	0.0030
month_9	126491	60403	43332026996	4.39	0.0451
ridership_lag1	0.41696	0.06464	4.111482E11	41.61	<.0001
gas_price_lag1	4616.66075	640.44100	5.134601E11	51.96	<.0001

Bounds on condition number: 1.4238, 19.518

Stepwise Selection: Step 5

Variable month_10 Entered: R-Square = 0.9048 and C(p) = 0.8886

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	2.252955E12	4.50591E11	53.24	<.0001
Error	28	2.369875E11	8463840558		
Corrected Total	33	2.489942E12			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-222361	69412	86859571044	10.26	0.0034
month_4	-179032	56728	84300815080	9.96	0.0038
month_9	141710	56256	53707176701	6.35	0.0178
month_10	138769	57343	49566822113	5.86	0.0223
ridership_lag1	0.38849	0.06097	3.436144E11	40.60	<.0001
gas_price_lag1	4823.03513	598.83534	5.490267E11	64.87	<.0001

Bounds on condition number: 1.4784, 30.285

All variables left in the model are significant at the 0.1500 level.

No other variable met the 0.0500 significance level for entry into the model.

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	gas_price_lag1		1	0.6685	0.6685	49.7122	64.54	<.0001
2	ridership_lag1		2	0.1515	0.8200	15.2762	26.10	<.0001
3	month_4		3	0.0475	0.8675	5.8609	10.75	0.0026
4	month_9		4	0.0174	0.8849	3.6758	4.39	0.0451
5	month_10		5	0.0199	0.9048	0.8886	5.86	0.0223

Pre-Covid PROC REG STEP Model 3.2

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

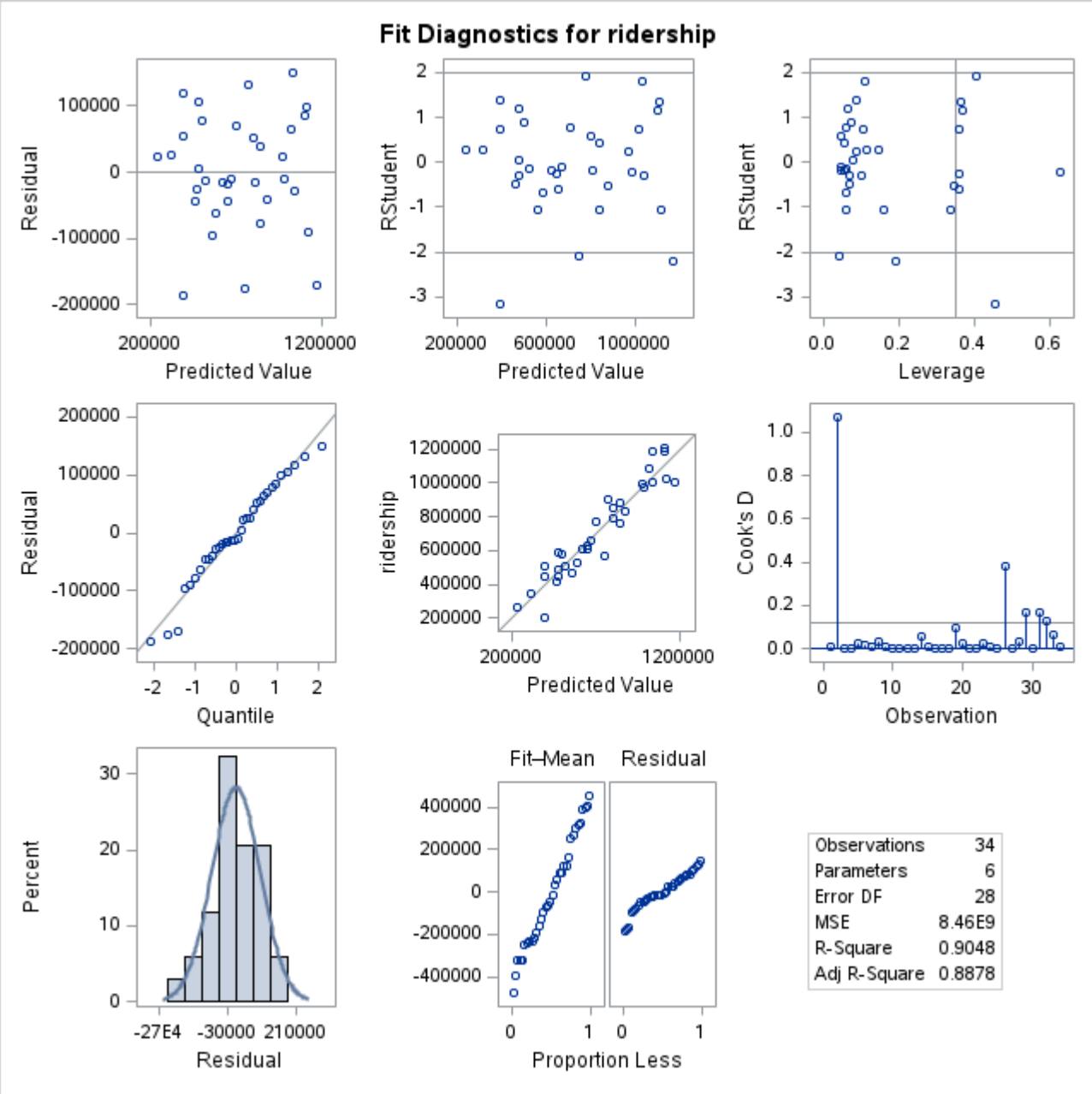
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	2.252955E12	4.50591E11	53.24	<.0001
Error	28	2.369875E11	8463840558		
Corrected Total	33	2.489942E12			

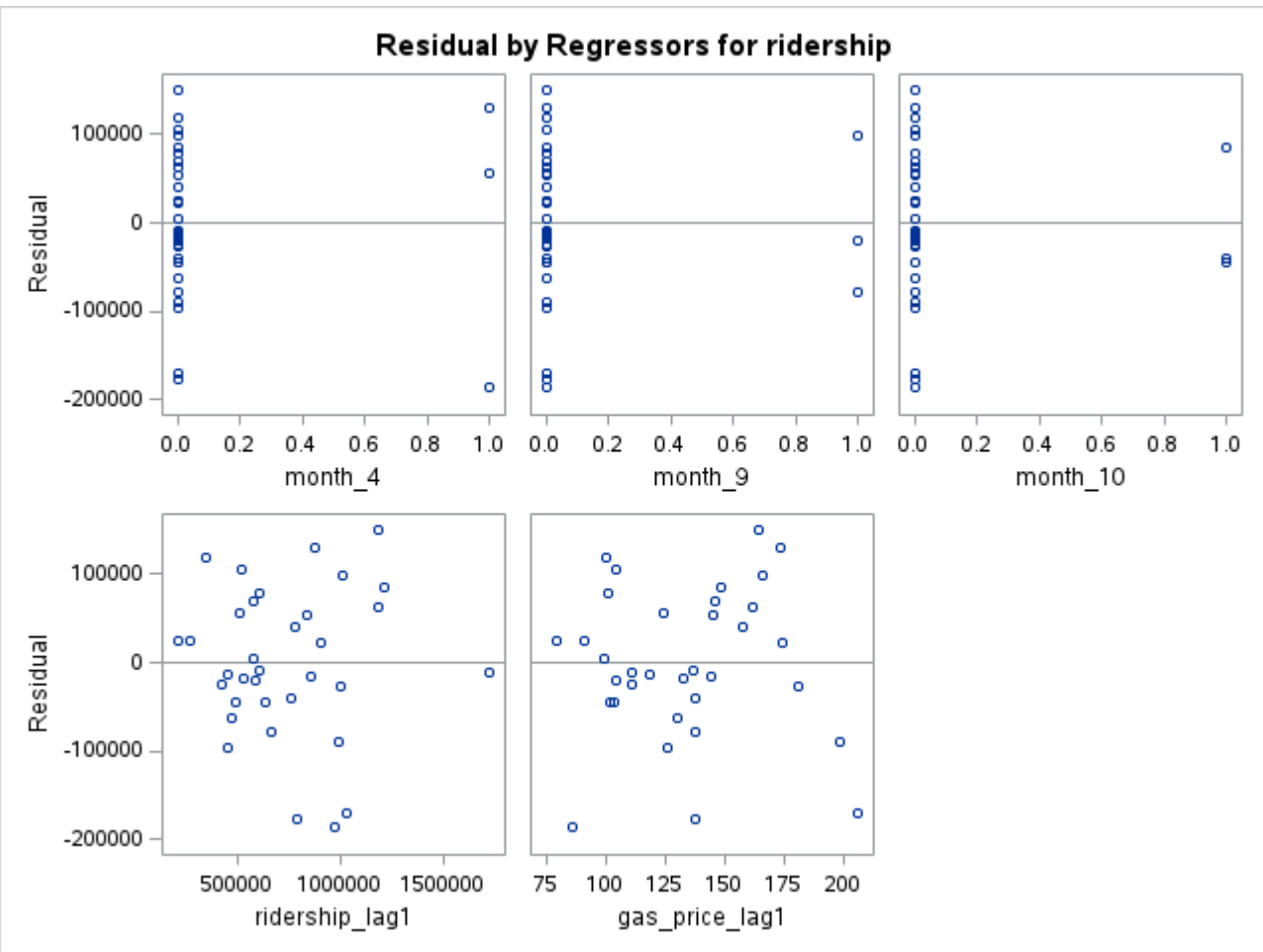
Root MSE	91999	R-Square	0.9048
Dependent Mean	715471	Adj R-Sq	0.8878
Coeff Var	12.85855		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	-222361	69412	-3.20	0.0034	.	0
month_4	1	-179032	56728	-3.16	0.0038	0.96154	1.04000
month_9	1	141710	56256	2.52	0.0178	0.97775	1.02276
month_10	1	138769	57343	2.42	0.0223	0.94103	1.06267
ridership_lag1	1	0.38849	0.06097	6.37	<.0001	0.67641	1.47839
gas_price_lag1	1	4823.03513	598.83534	8.05	<.0001	0.68810	1.45327

Pre-Covid PROC REG STEP Model 3.2

The REG Procedure
Model: MODEL1
Dependent Variable: ridership





Pre-COVID Model P1: Month-only Seasonal Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	5.662673E11	51478845811	0.59	0.8178
Error	22	1.923675E12	87439780303		
Corrected Total	33	2.489942E12			

Root MSE	295702	**R-Square**	0.2274

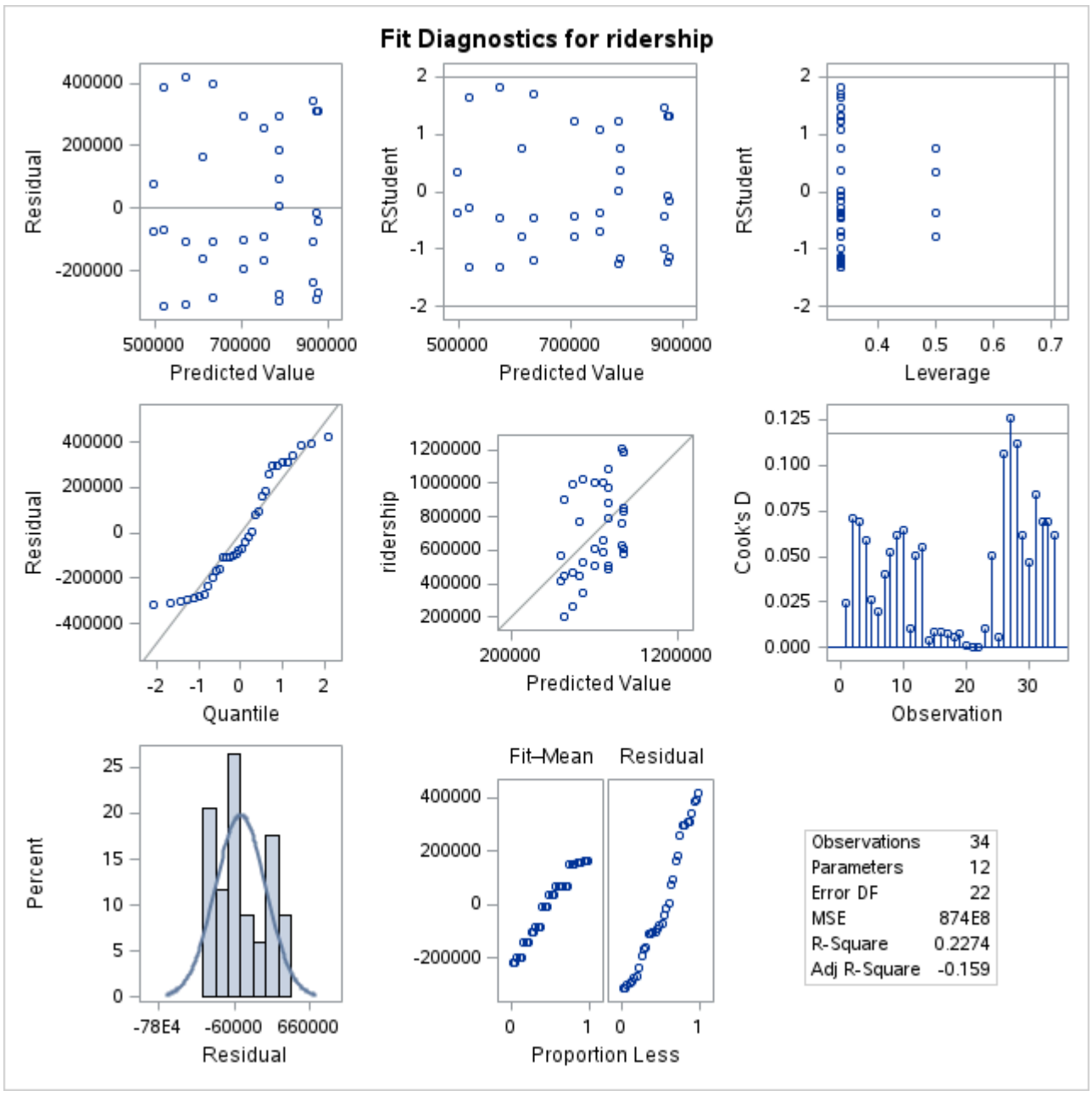
Dependent Mean	715471	**Adj R-Sq**	-0.1589
Coeff Var	41.32975		

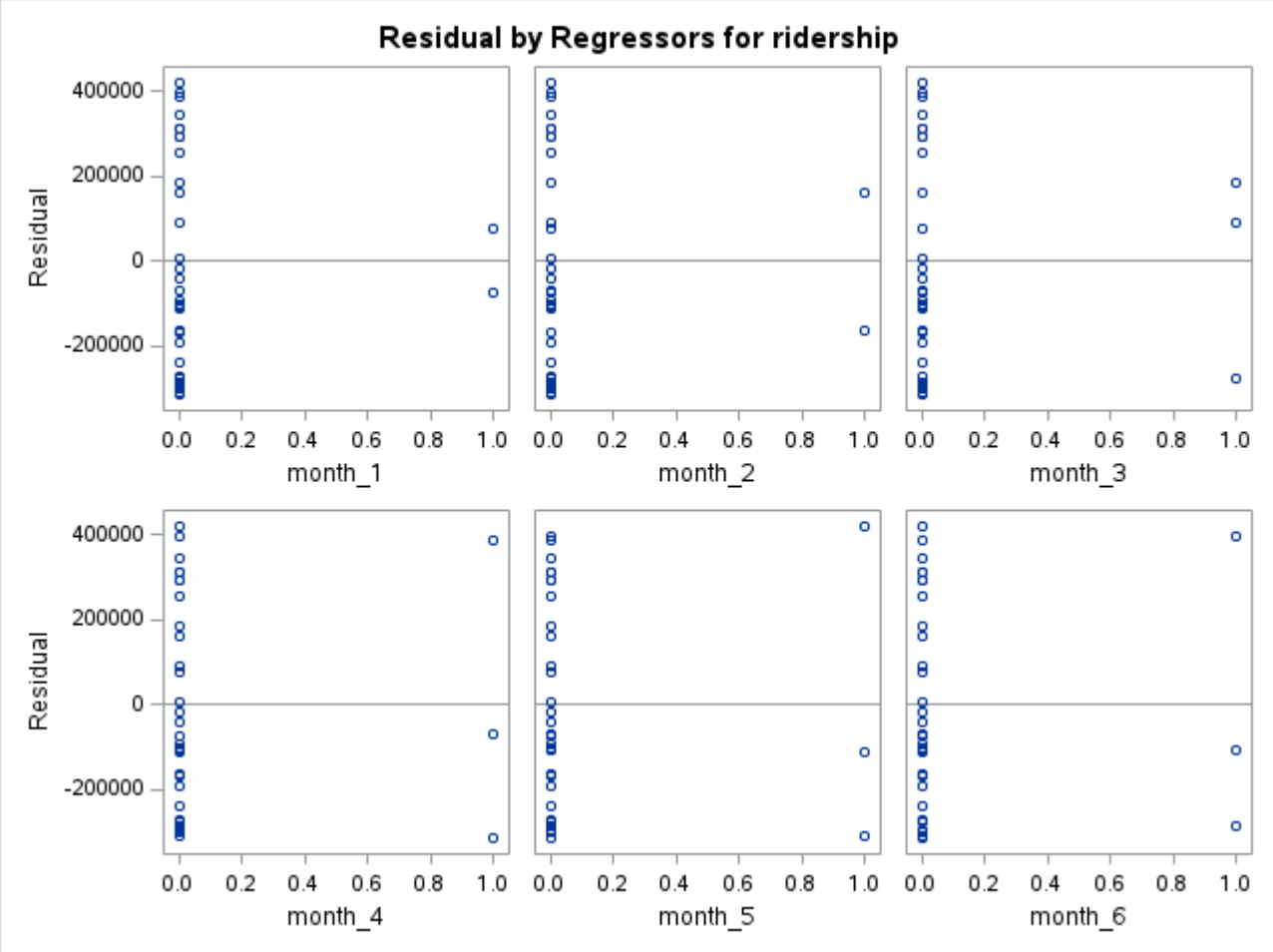
Parameter Estimates

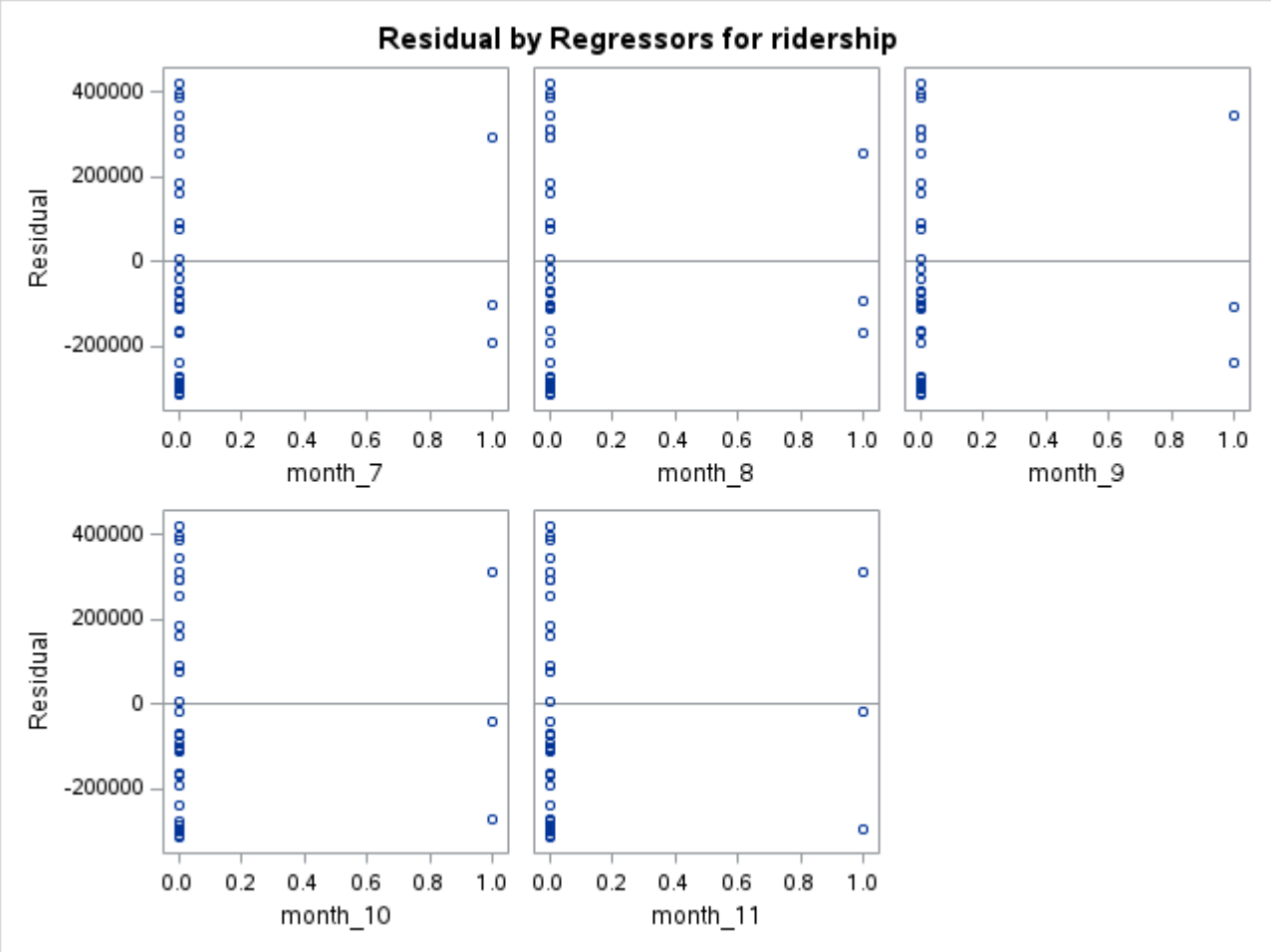
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	785333	170724	4.60	0.0001	.	0
month_1	1	-288833	269938	-1.07	0.2962	0.63750	1.56863
month_2	1	-172333	269938	-0.64	0.5298	0.63750	1.56863
month_3	1	1000.00000	241440	0.00	0.9967	0.54839	1.82353
month_4	1	-265667	241440	-1.10	0.2831	0.54839	1.82353
month_5	1	-212000	241440	-0.88	0.3894	0.54839	1.82353
month_6	1	-152667	241440	-0.63	0.5337	0.54839	1.82353
month_7	1	-80000	241440	-0.33	0.7435	0.54839	1.82353
month_8	1	-33333	241440	-0.14	0.8914	0.54839	1.82353
month_9	1	80000	241440	0.33	0.7435	0.54839	1.82353
month_10	1	91667	241440	0.38	0.7078	0.54839	1.82353
month_11	1	86667	241440	0.36	0.7230	0.54839	1.82353

Pre-COVID Model P1: Month-only Seasonal Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership







Pre-COVID Model P2: Lag Memory Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1.697893E12	8.489466E11	33.23	<.0001
Error	31	7.920493E11	25549978977		
Corrected Total	33	2.489942E12			

Root MSE	159844	**R-Square**	0.6819
Dependent Mean	715471	**Adj R-Sq**	0.6614
Coeff Var	22.34105		

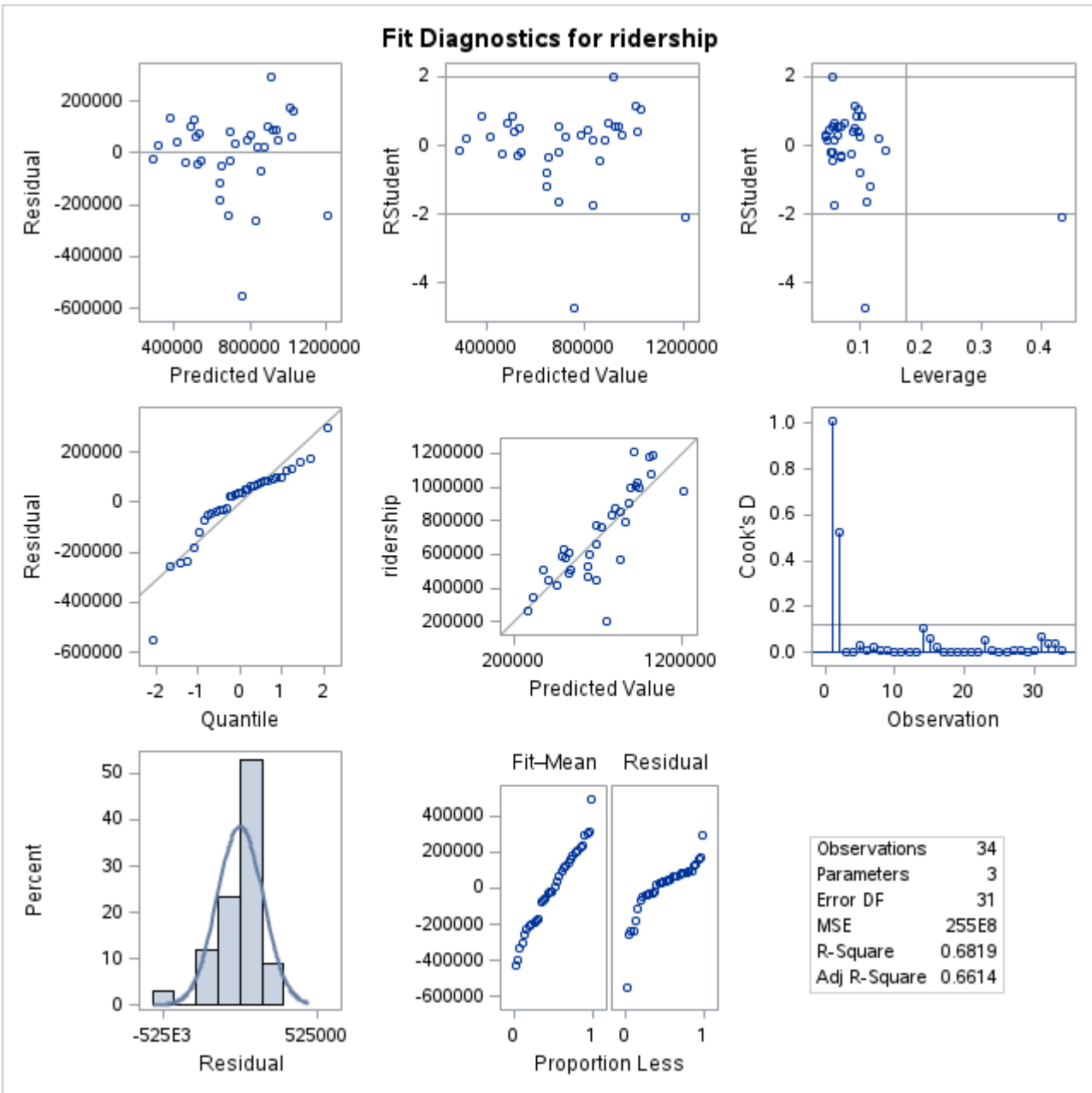
Parameter Estimates

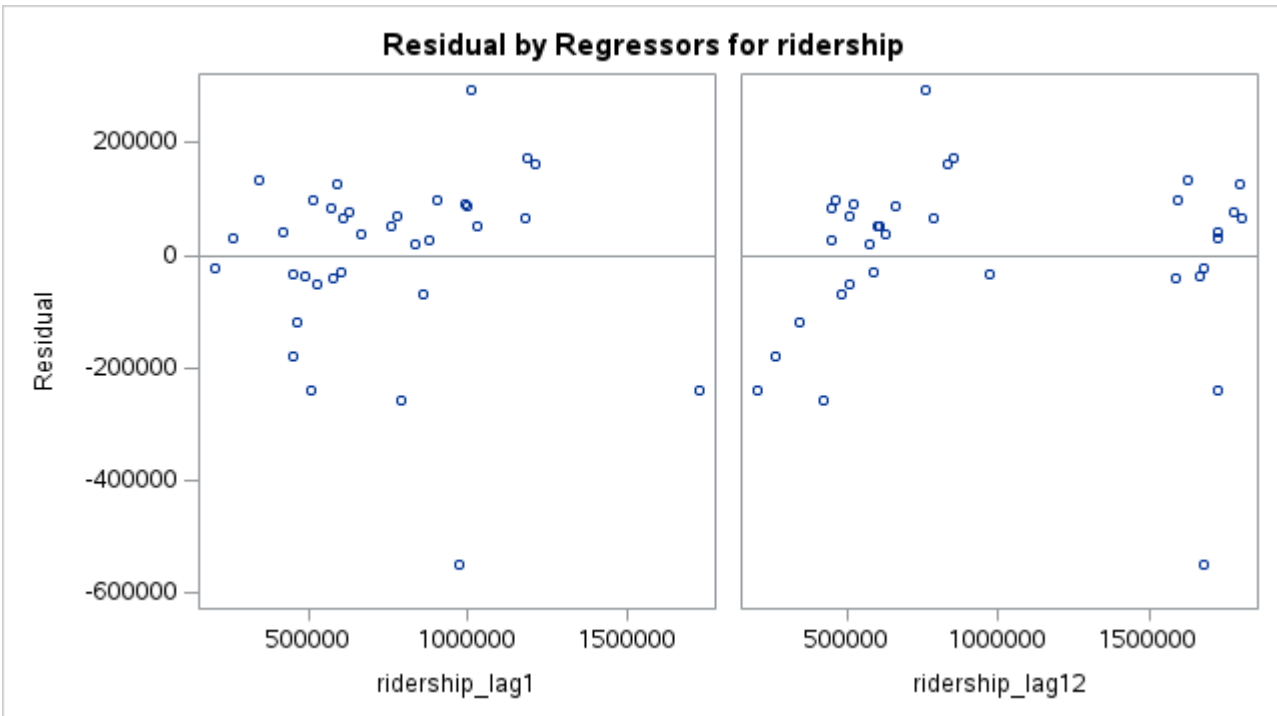
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	408785	91887	4.45	0.0001	.	0
ridership_lag1	1	0.61225	0.08865	6.91	<.0001	0.96595	1.03526

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
ridership_lag12	1	-0.14773	0.04955	-2.98	0.0055	0.96595	1.03526

Pre-COVID Model P2: Lag Memory Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership





Pre-COVID Model P3: Reduced External Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	2.103436E12	5.258589E11	39.46	<.0001

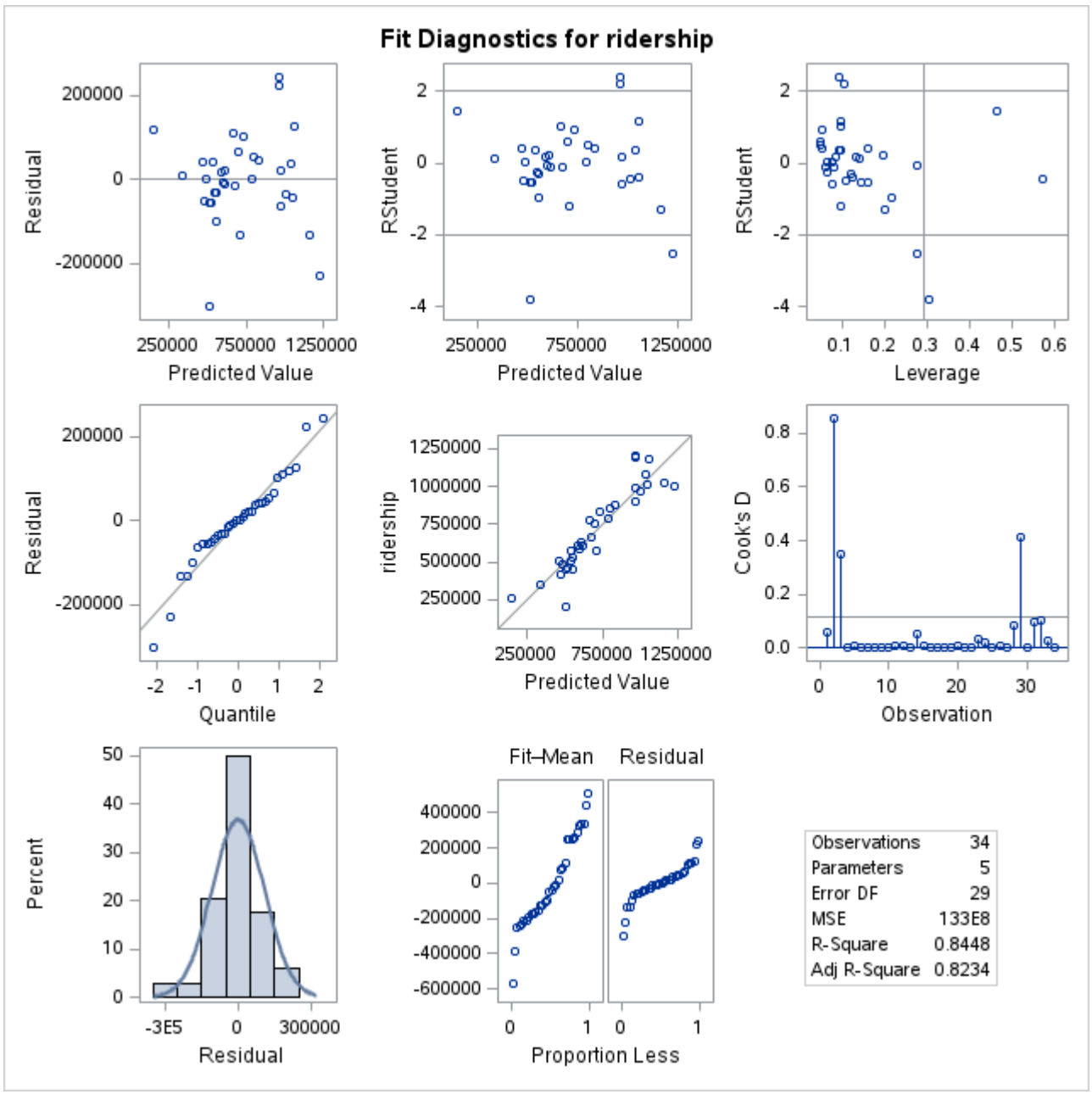
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Error	29	3.86507E11	13327826100		
Corrected Total	33	2.489942E12			

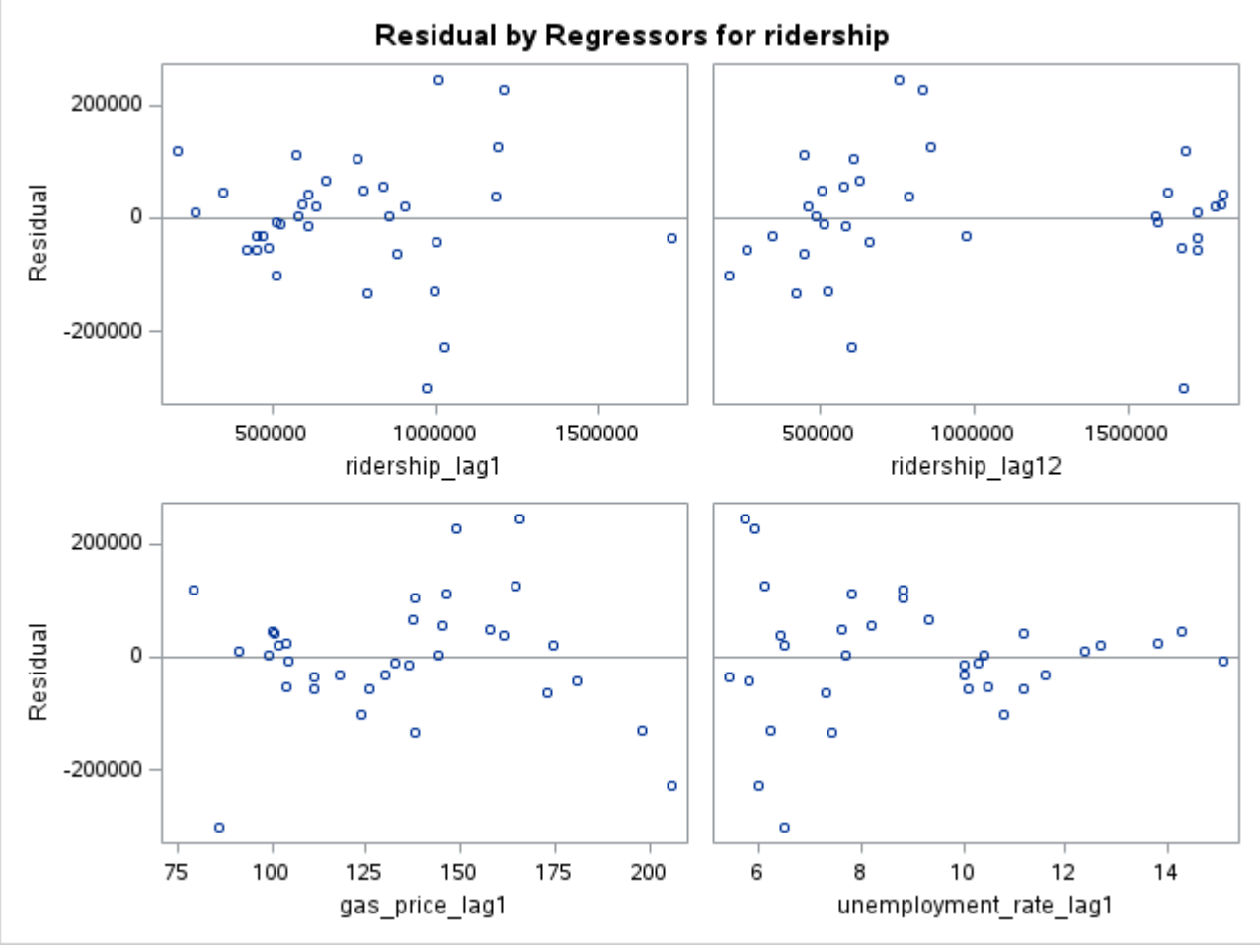
Root MSE	115446	R-Square	0.8448
Dependent Mean	715471	Adj R-Sq	0.8234
Coeff Var	16.13570		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	-738581	254759	-2.90	0.0071	.	0
ridership_lag1	1	0.48653	0.11177	4.35	0.0002	0.31696	3.15496
ridership_lag12	1	0.04711	0.05814	0.81	0.4244	0.36592	2.73287
gas_price_lag1	1	6298.35822	1158.95146	5.43	<.0001	0.28929	3.45677
unemployment_rate_lag1	1	23274	14003	1.66	0.1073	0.28107	3.55788

Pre-COVID Model P3: Reduced External Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership





Pre-Covid Model P4: Whithout Ridership History

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	34
Number of Observations Used	34

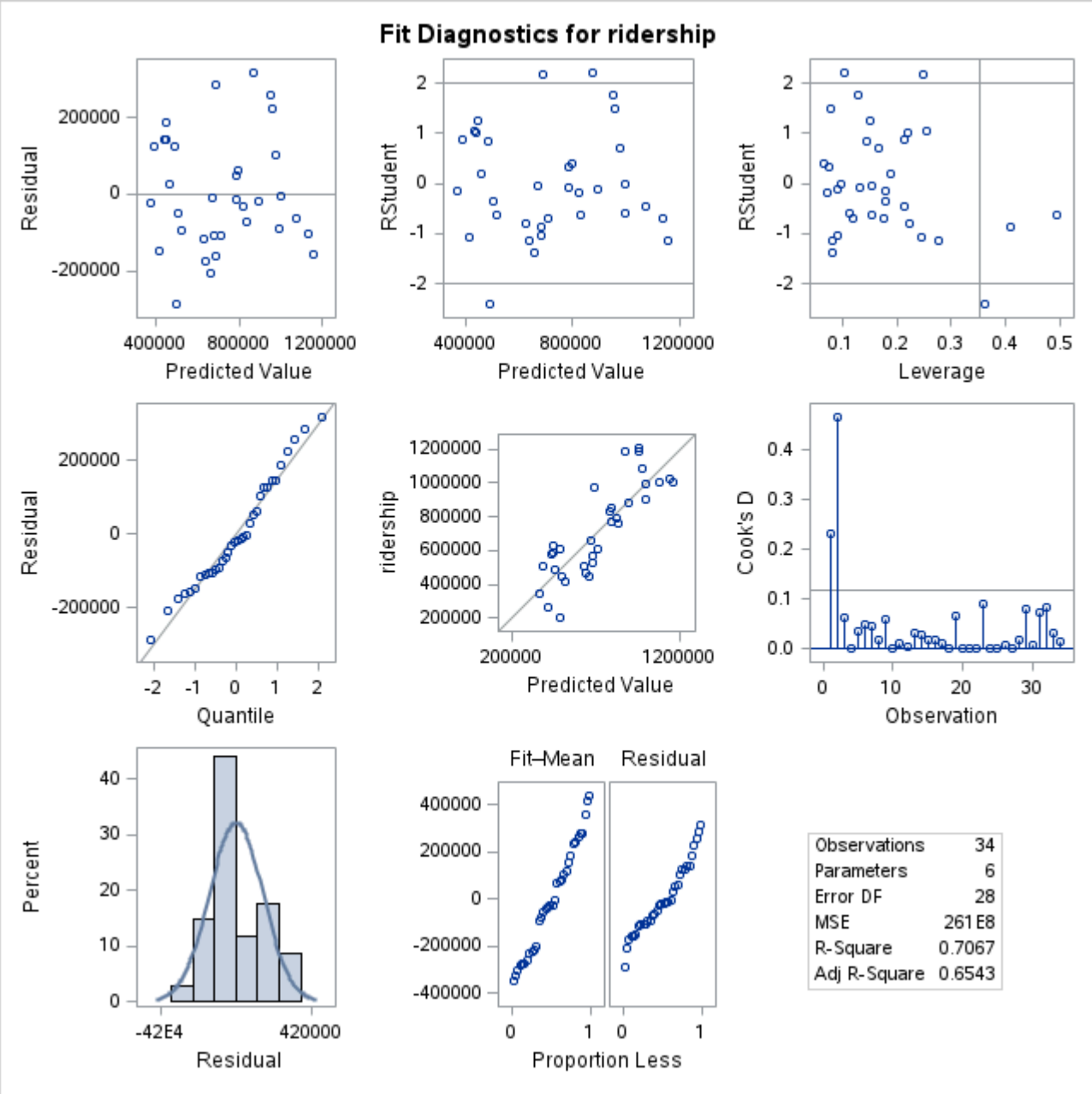
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1.75961E12	3.51922E11	13.49	<.0001
Error	28	7.303326E11	26083306505		
Corrected Total	33	2.489942E12			

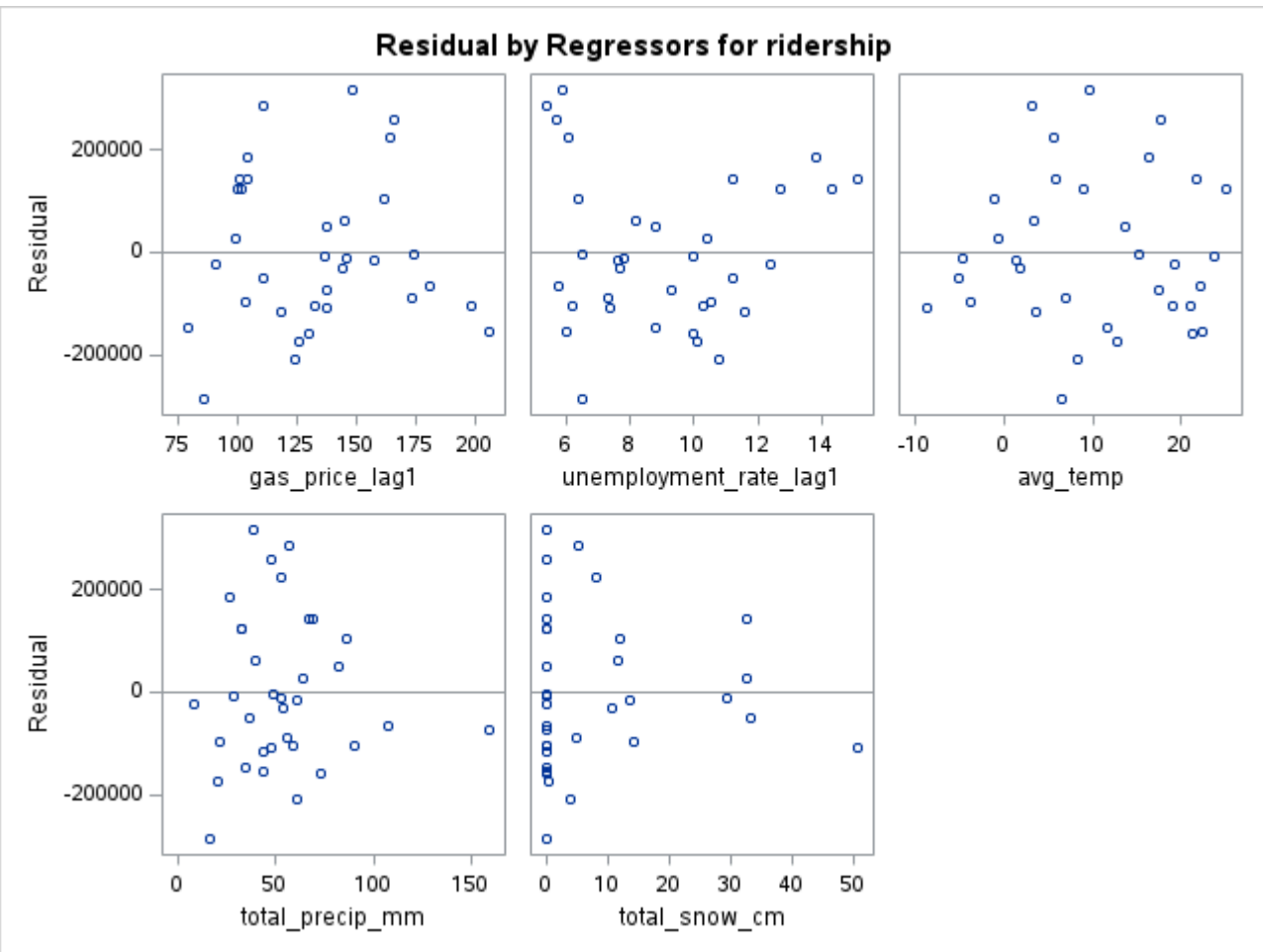
Root MSE	161503	R-Square	0.7067
Dependent Mean	715471	Adj R-Sq	0.6543
Coeff Var	22.57301		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	119369	278462	0.43	0.6714	.	0
gas_price_lag1	1	5725.07417	1334.52782	4.29	0.0002	0.42698	2.34203
unemployment_rate_lag1	1	-17049	16047	-1.06	0.2971	0.41881	2.38770
avg_temp	1	-3641.14481	5148.52398	-0.71	0.4853	0.32623	3.06532
total_precip_mm	1	941.64677	1055.83406	0.89	0.3801	0.85746	1.16623
total_snow_cm	1	-3472.08099	3460.59856	-1.00	0.3243	0.39073	2.55928

Pre-Covid Model P4: Whithout Ridership History

The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
Quantile
ridership
Cook's D
Observation
Percent
Fit–Mean
Proportion Less
Observations 34
Parameters 6
Error DF 28
MSE 261E8
R-Square 0.7067
Adj R-Square 0.6543


Residual by Regressors for ridership
Residual
gas_price_lag1
unemployment_rate_lag1
avg_temp
total_precip_mm
total_snow_cm